Introduction to deep learning

INTRODUCTION TO DEEP LEARNING IN PYTHON



Dan Becker

Data Scientist and contributor to Keras and TensorFlow libraries

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Imagine you work for a bank

- You need to predict how many transactions each customer will make next year

Example as seen by linear regression



Example as seen by linear regression



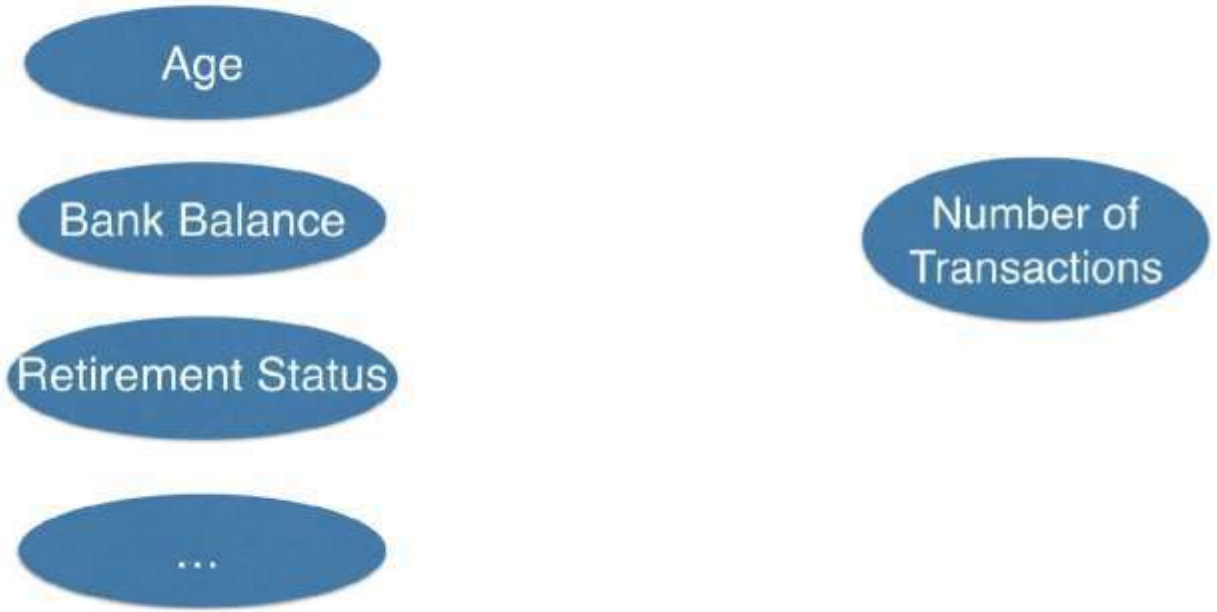
Example as seen by linear regression



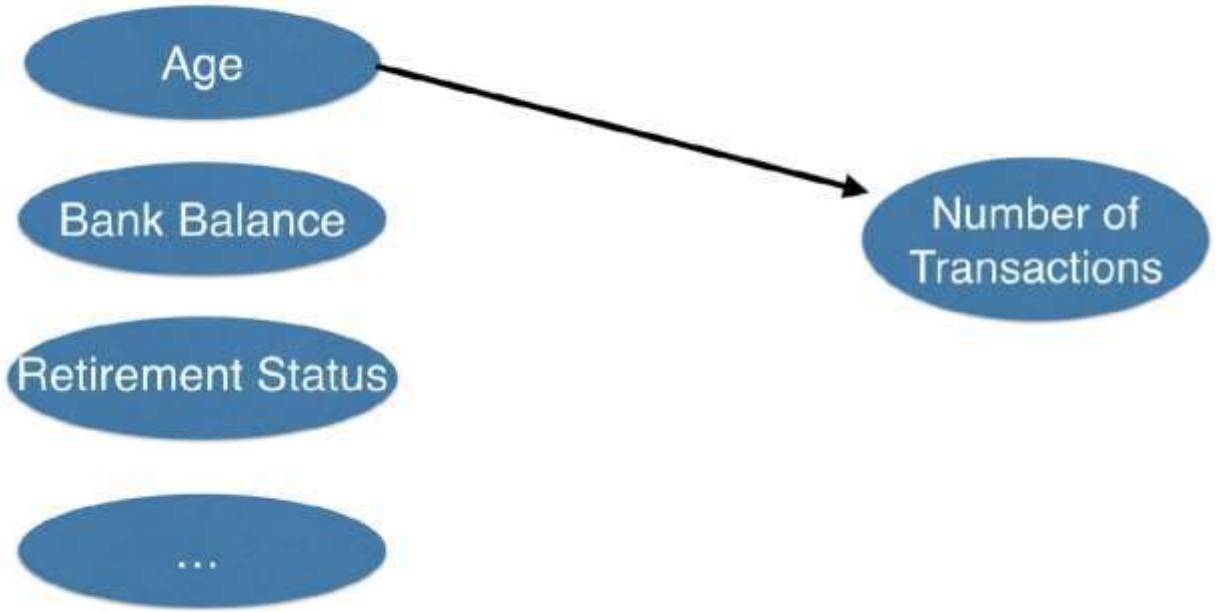
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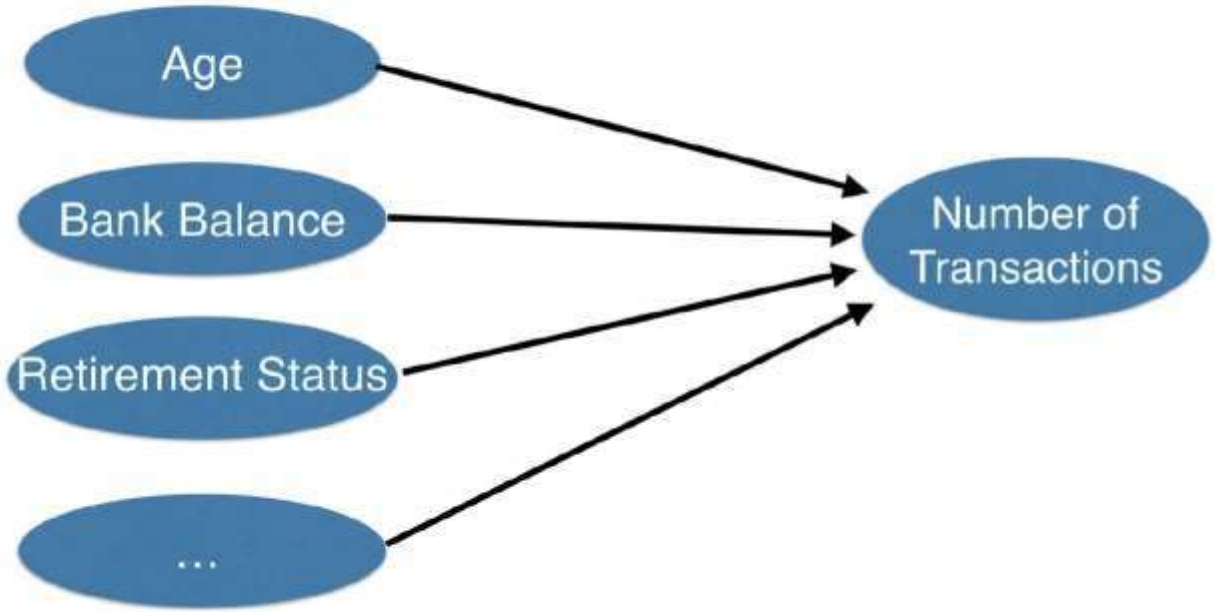
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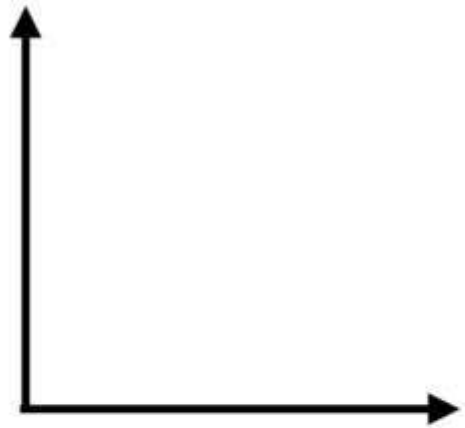
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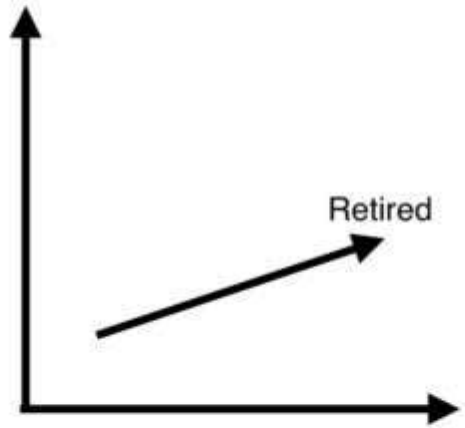
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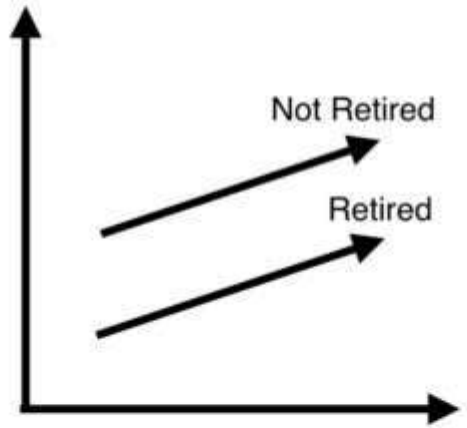
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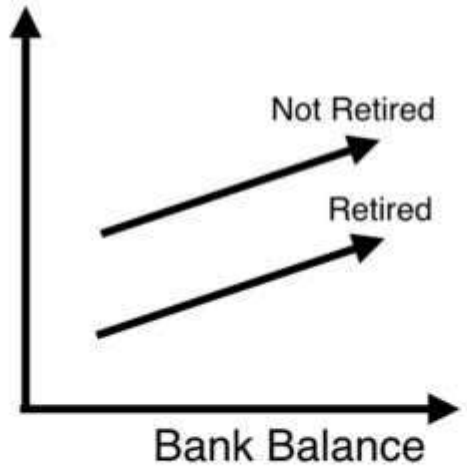
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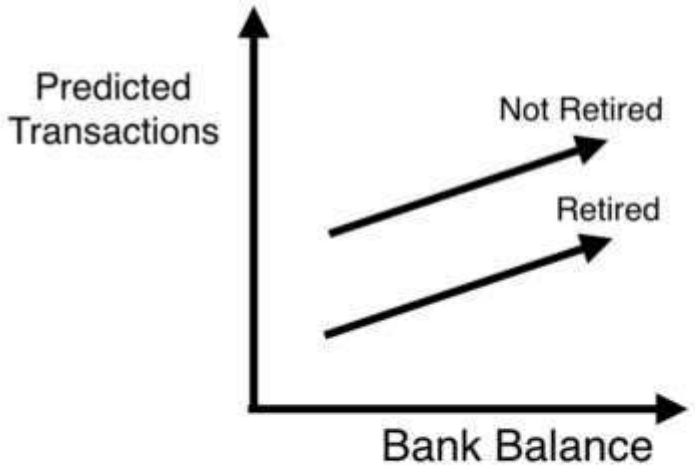
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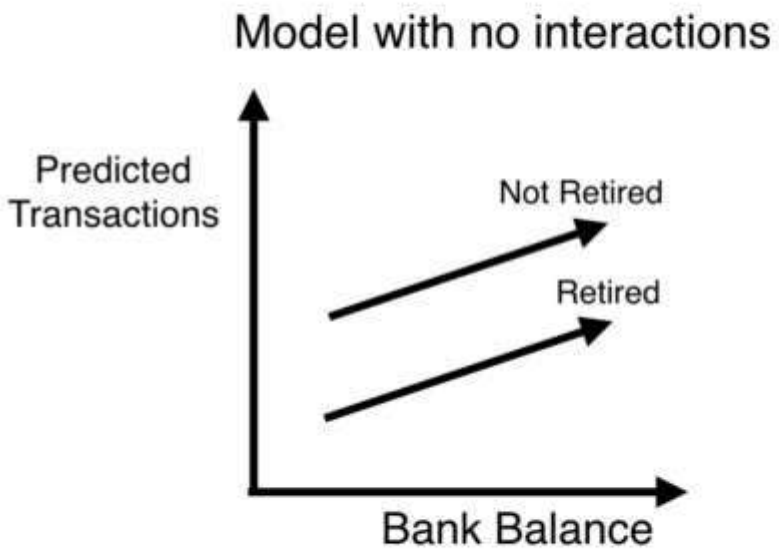
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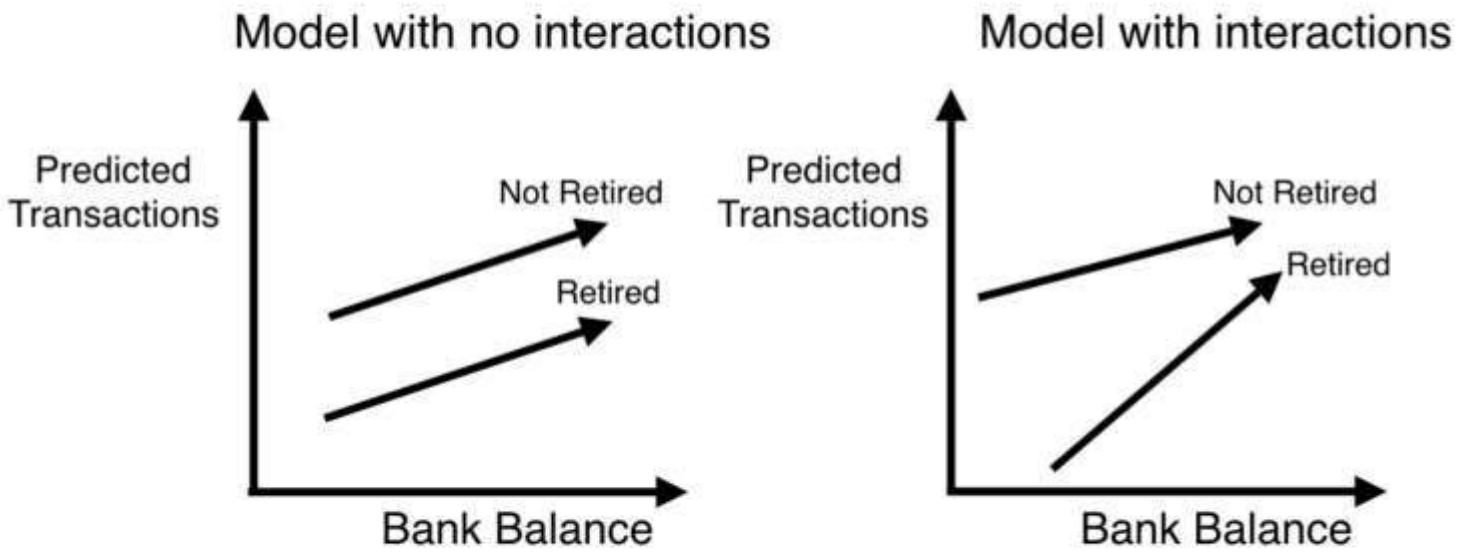
Example as seen by linear regression



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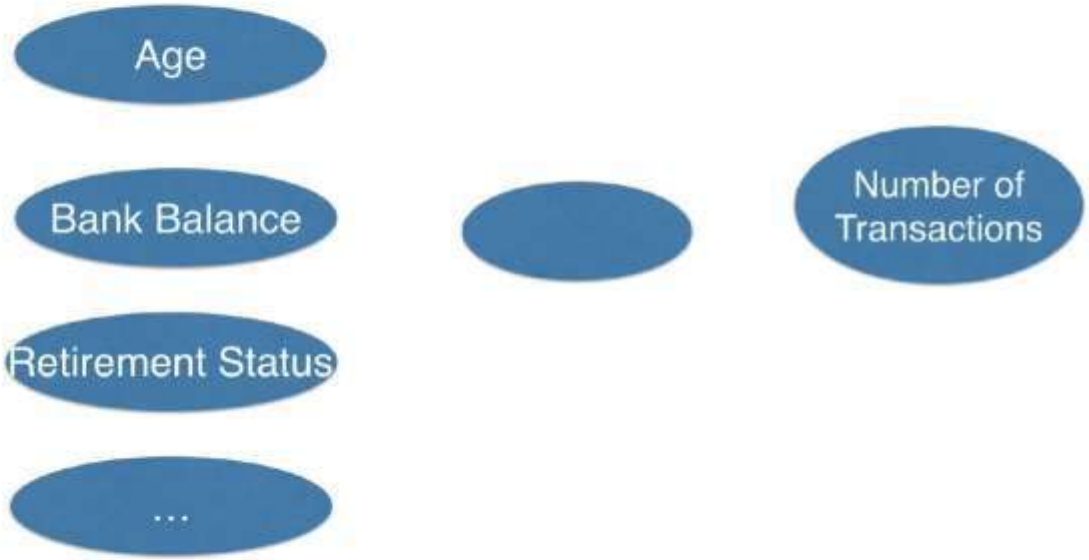
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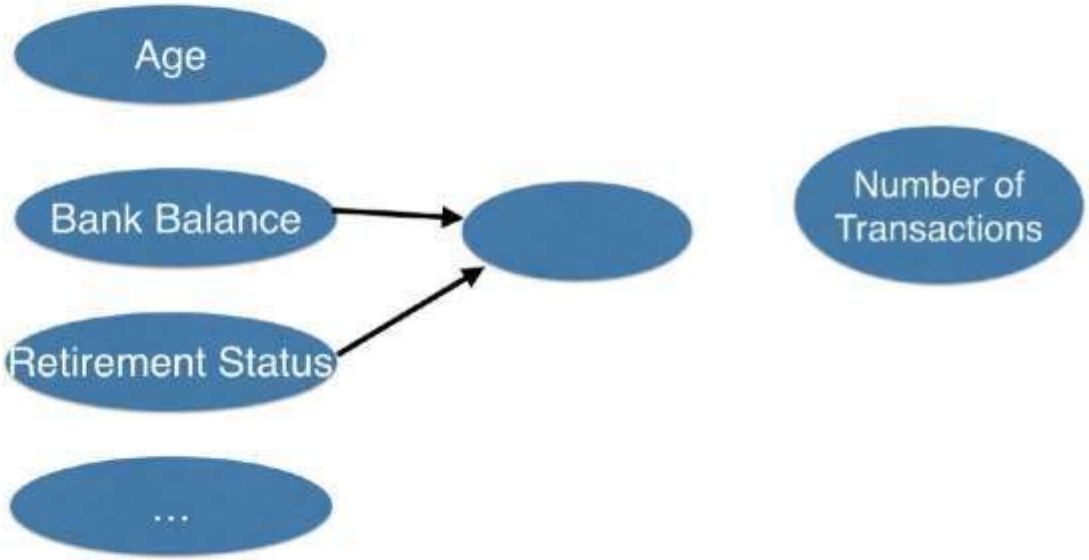
Interactions

- Neural networks account for interactions really well
- Deep learning uses especially powerful neural networks
  - Text
  - Images
  - Videos
  - Audio
  - Source code

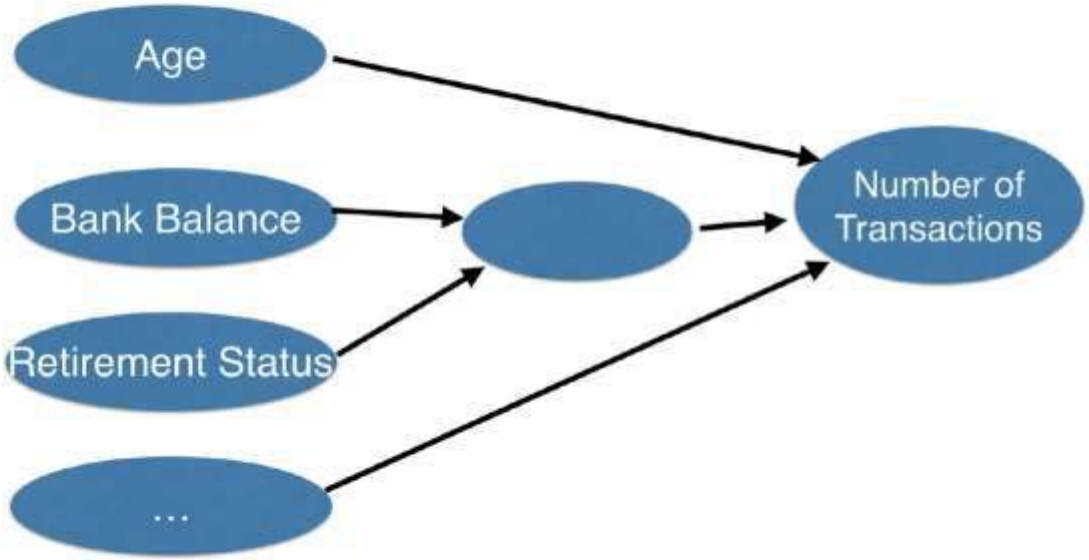
Deep learning models capture interactions



Deep learning models capture interactions



Deep learning models capture interactions



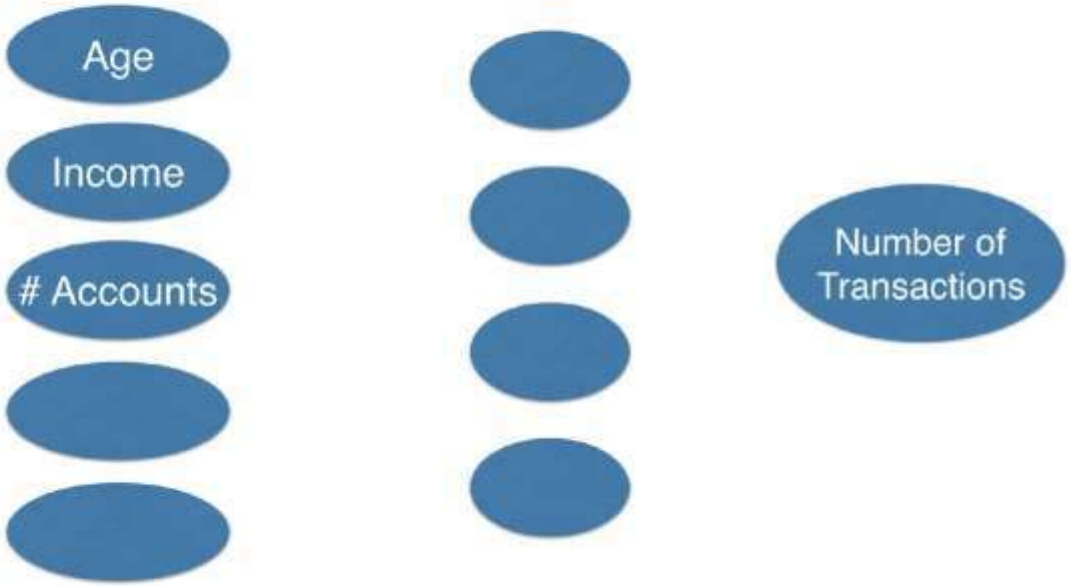
Interactions in neural network



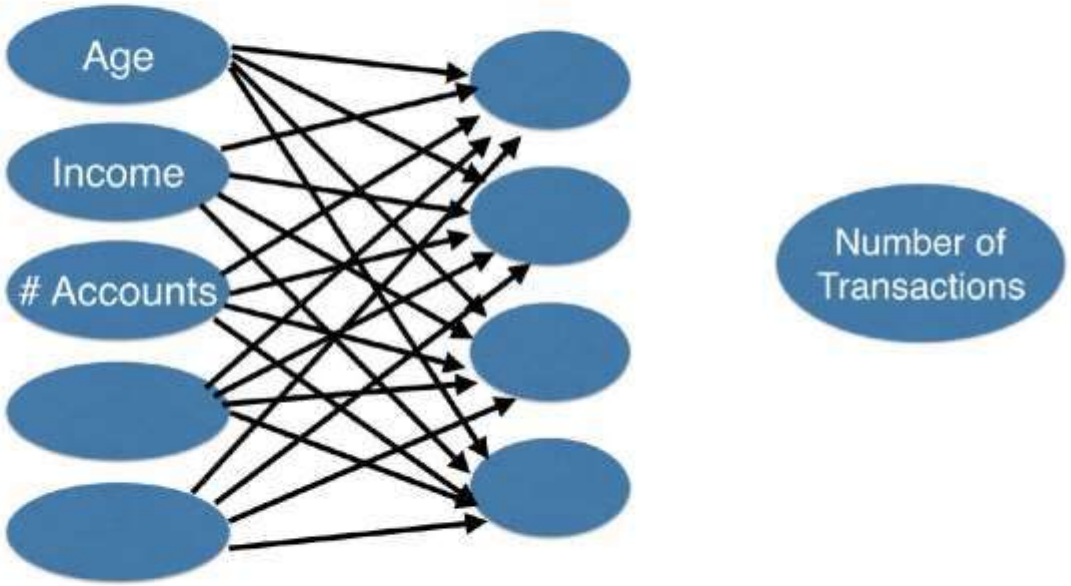
Interactions in neural network



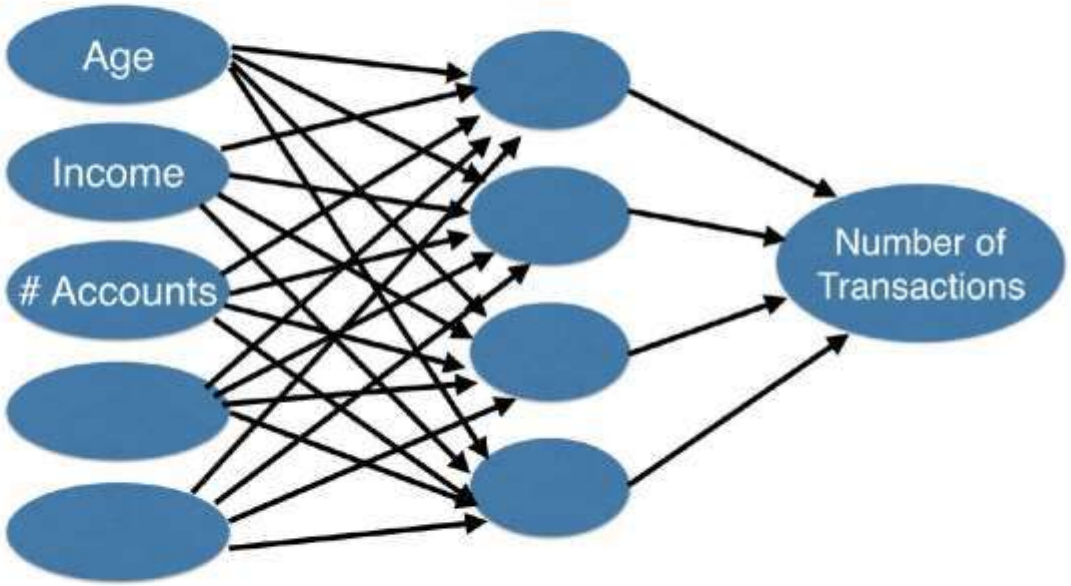
Interactions in neural network



Interactions in neural network



Interactions in neural network



Forward propagation

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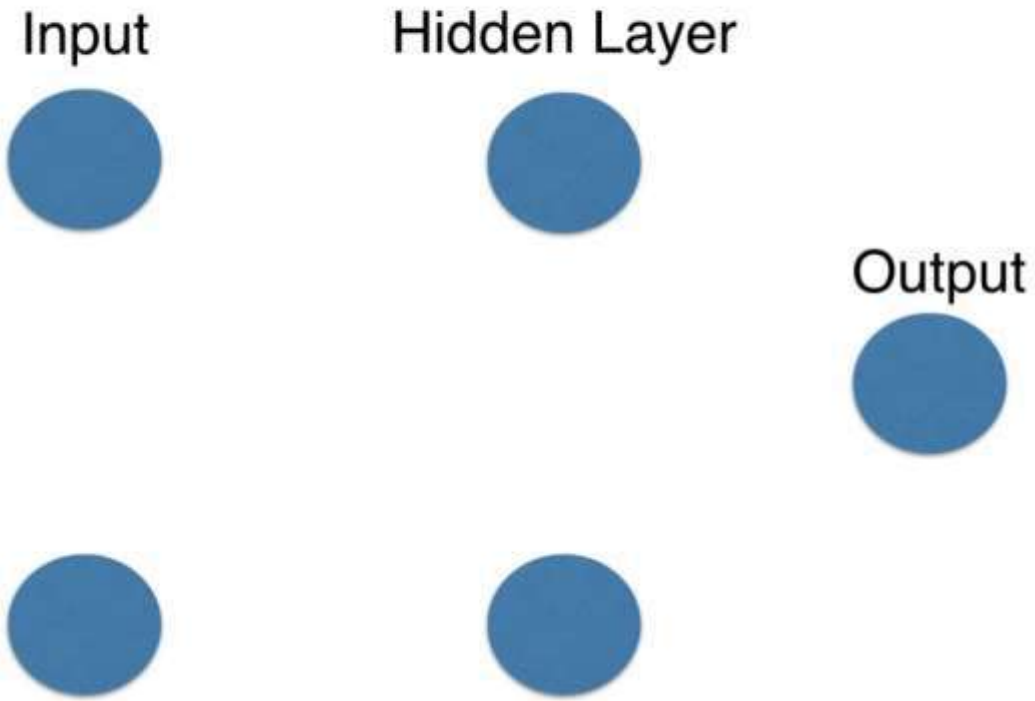
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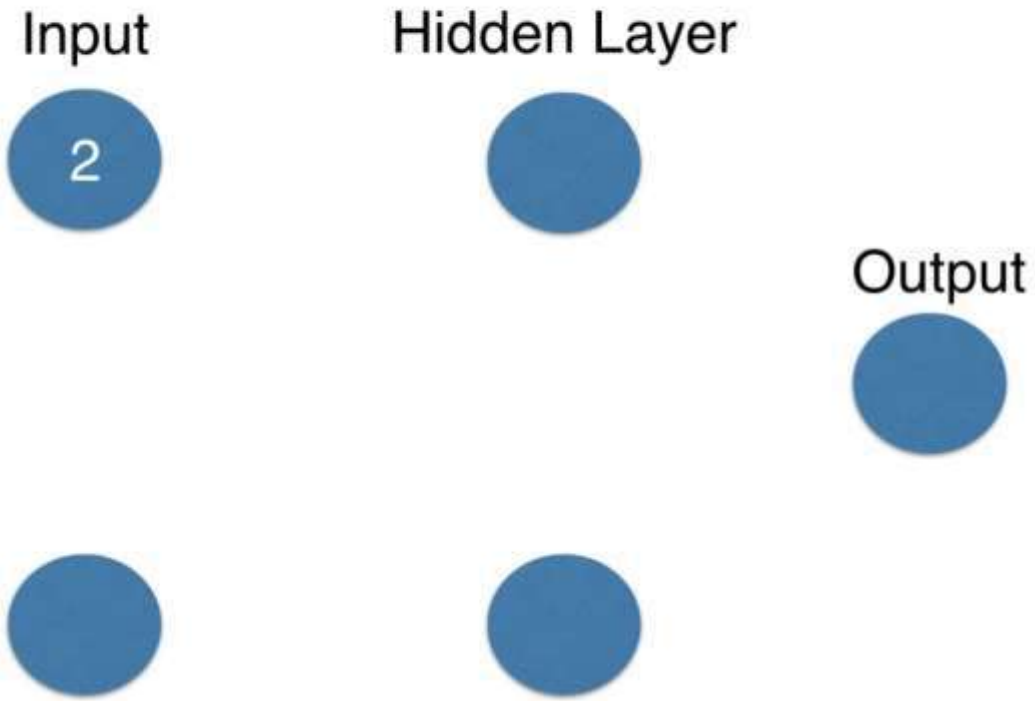
Bank transactions example

- Make predictions based on:
  - Number of children
  - Number of existing accounts

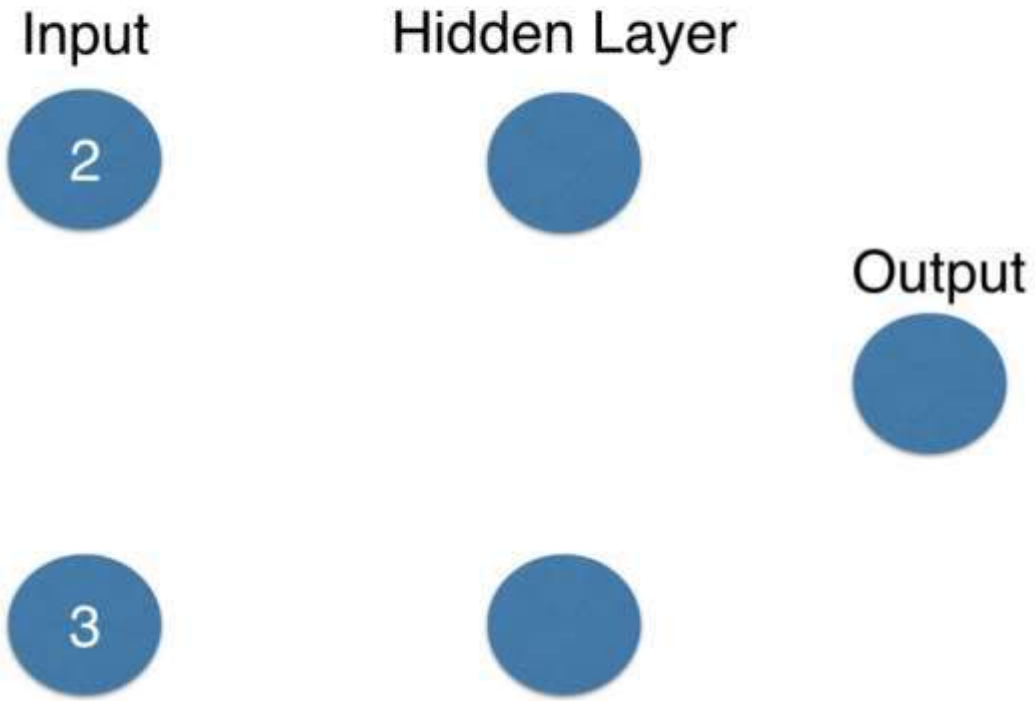
Forward propagation



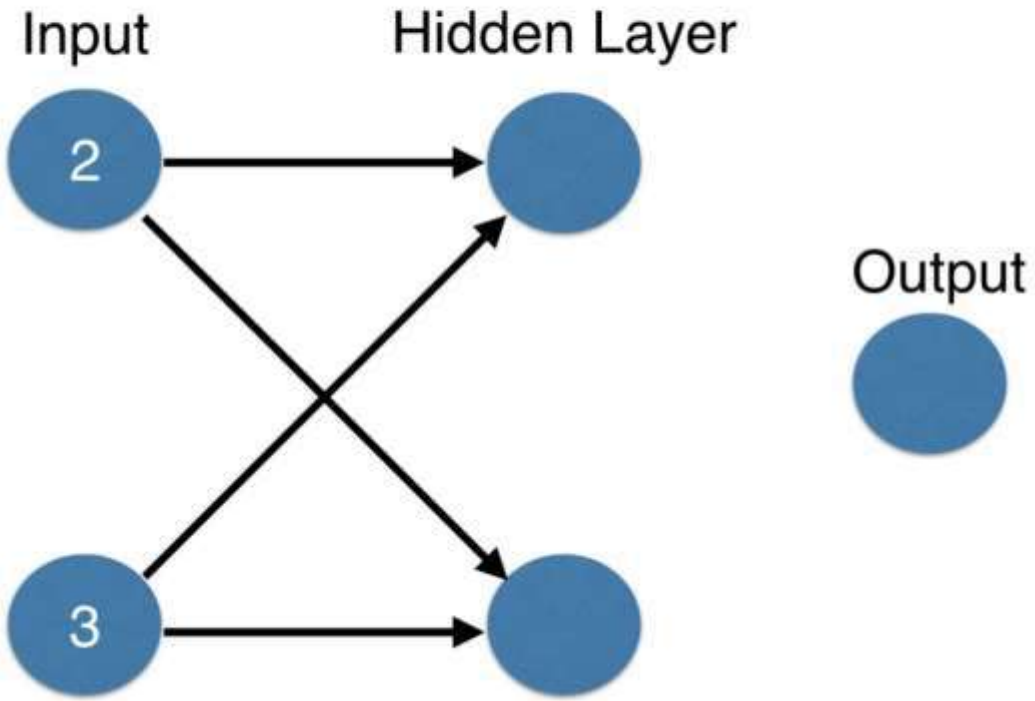
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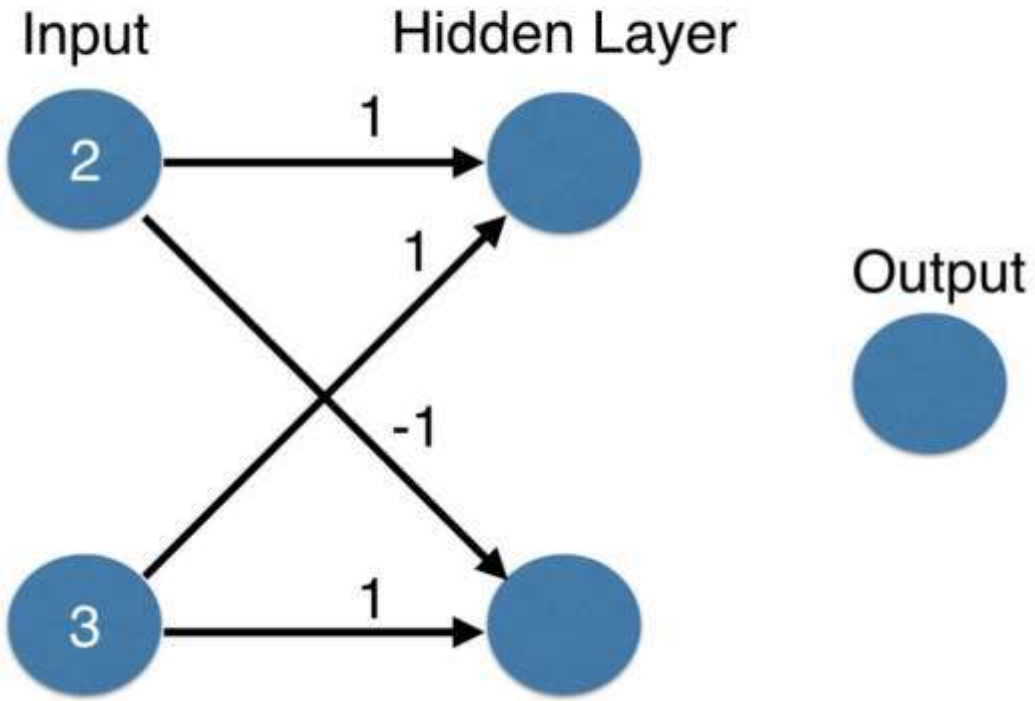
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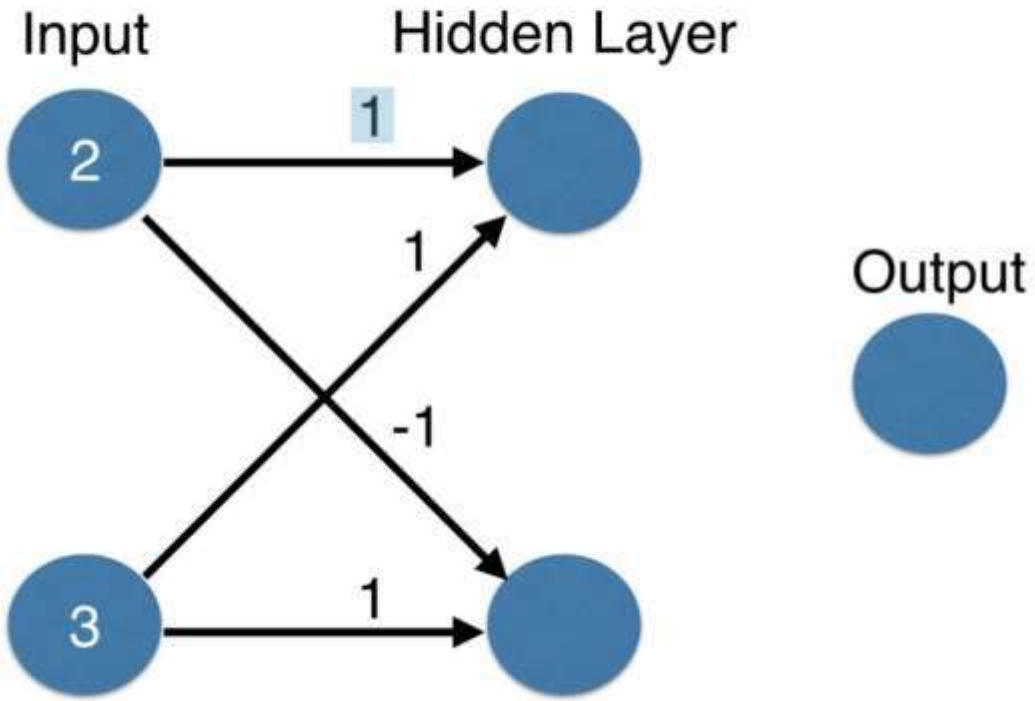
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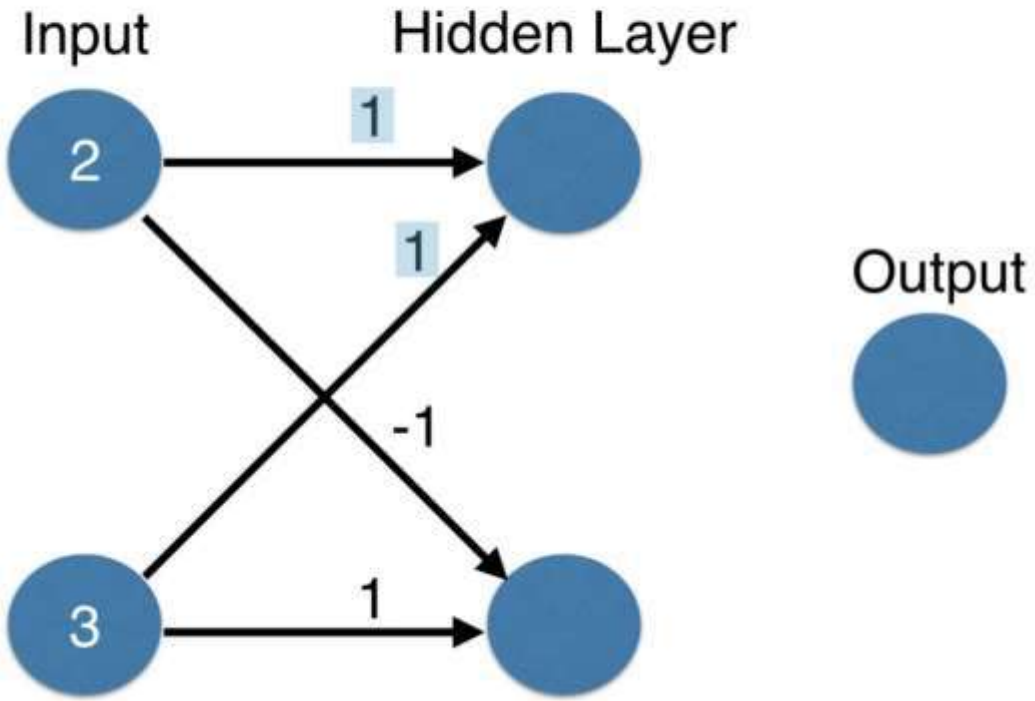
Forward propagation



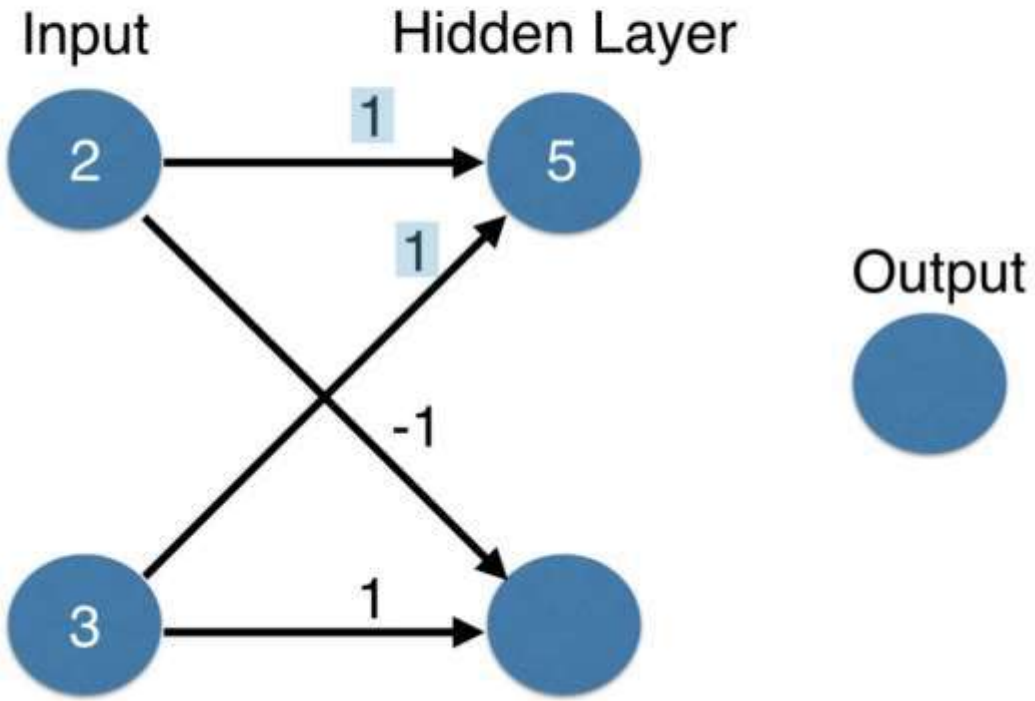
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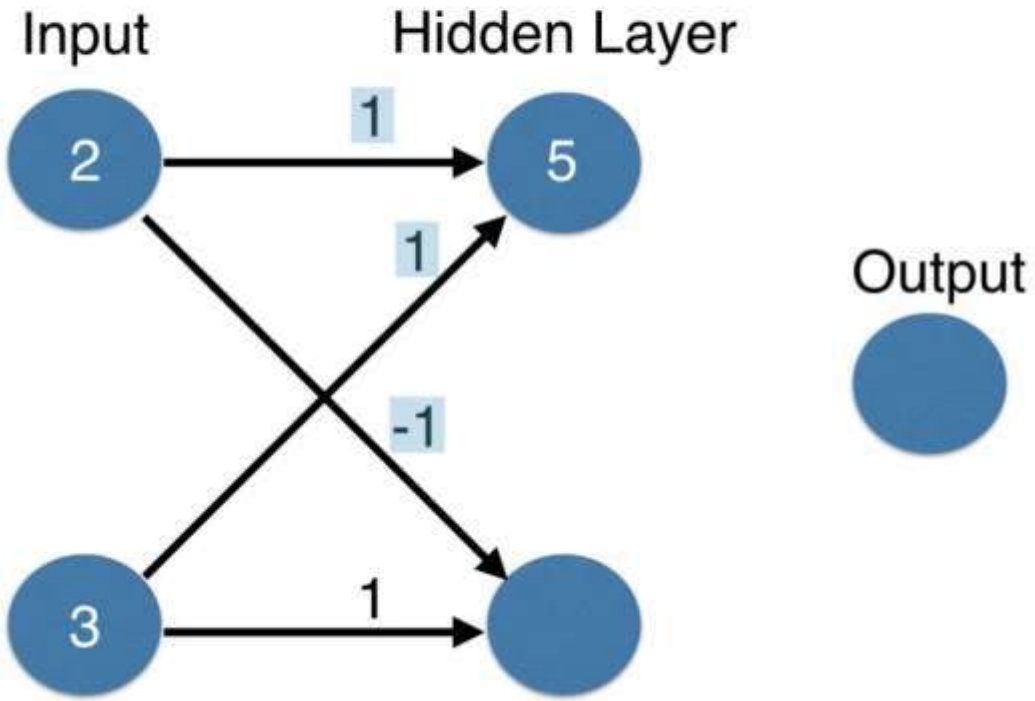
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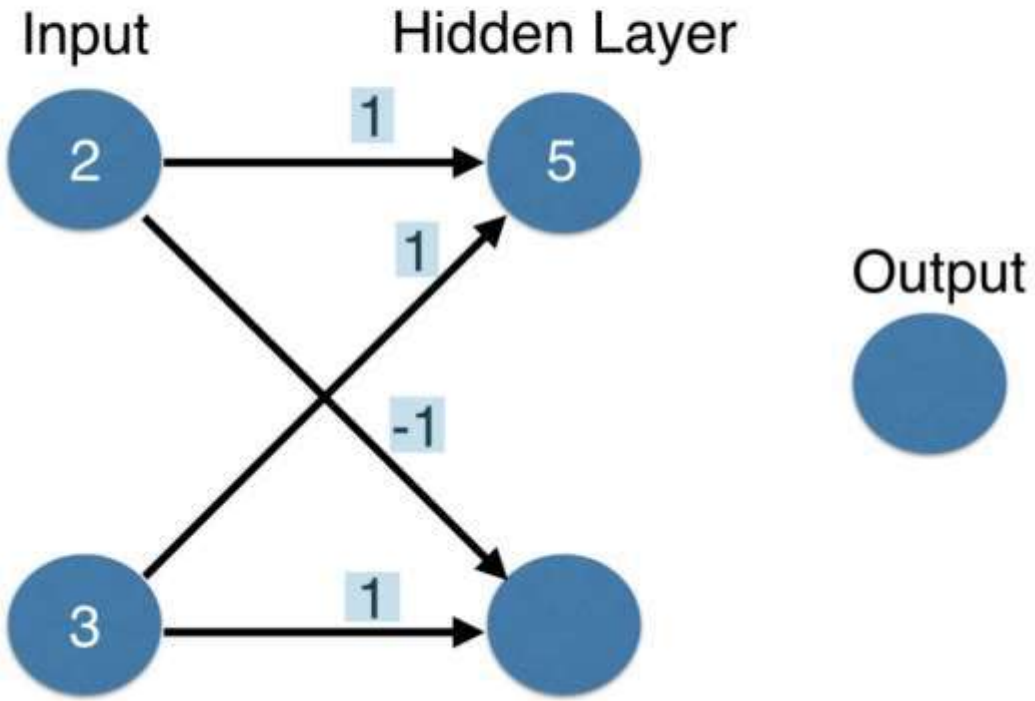
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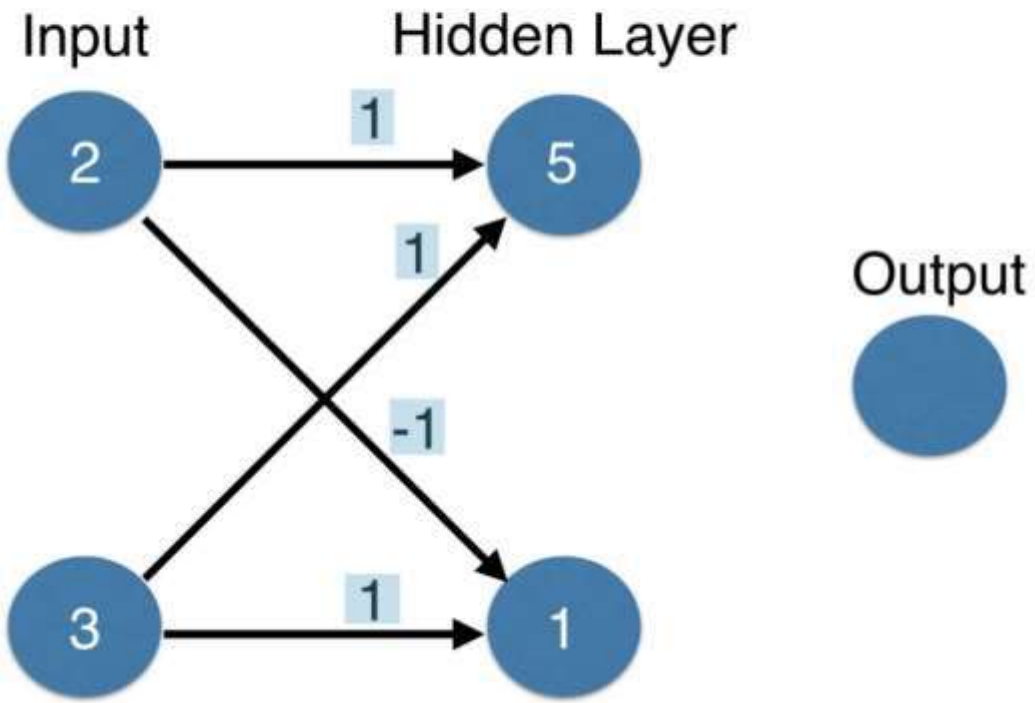
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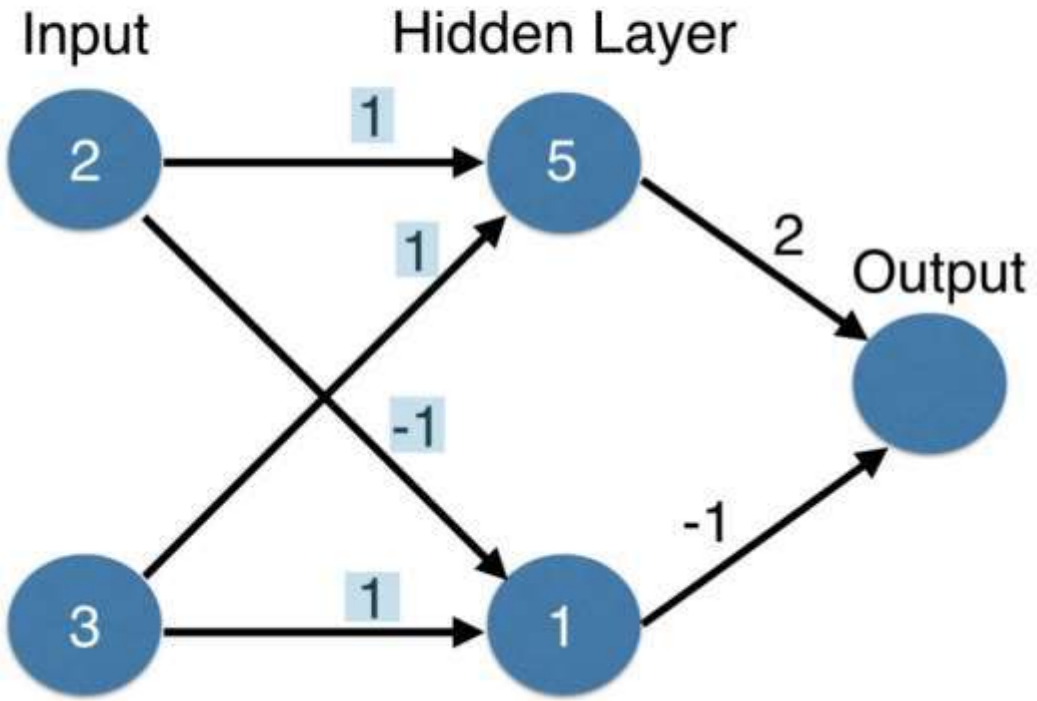
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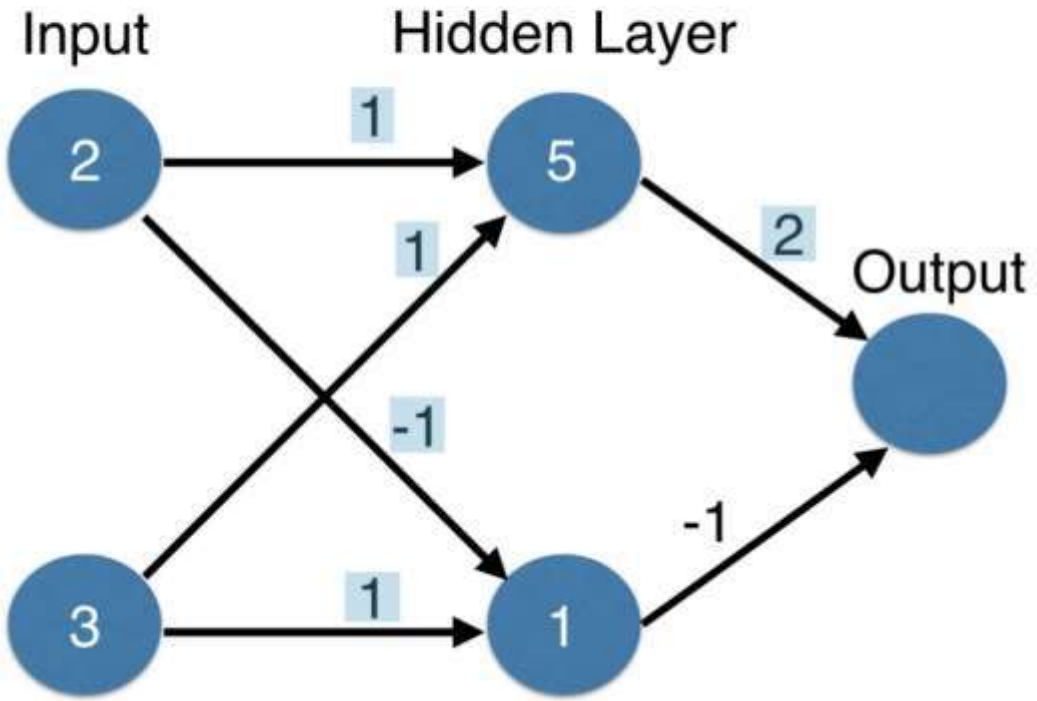
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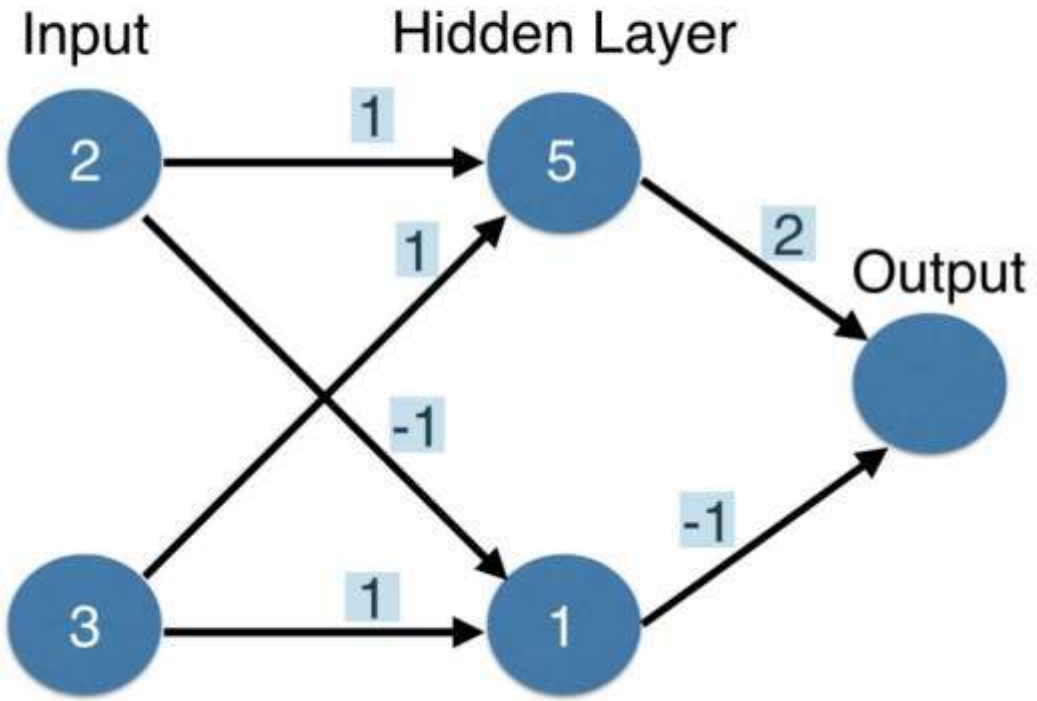
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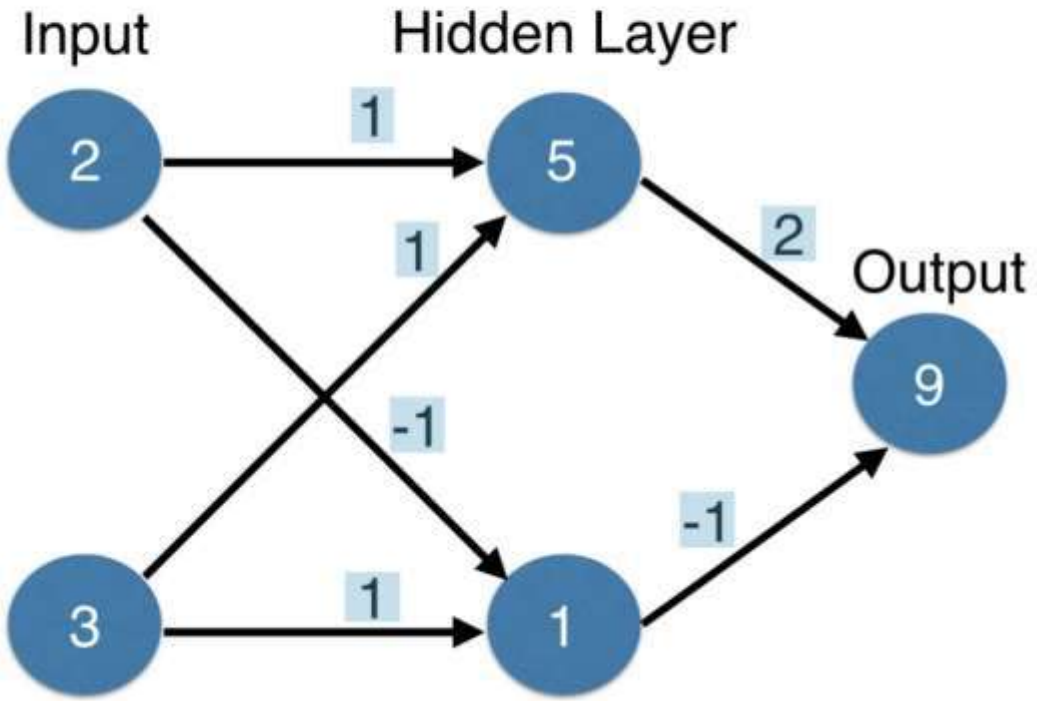
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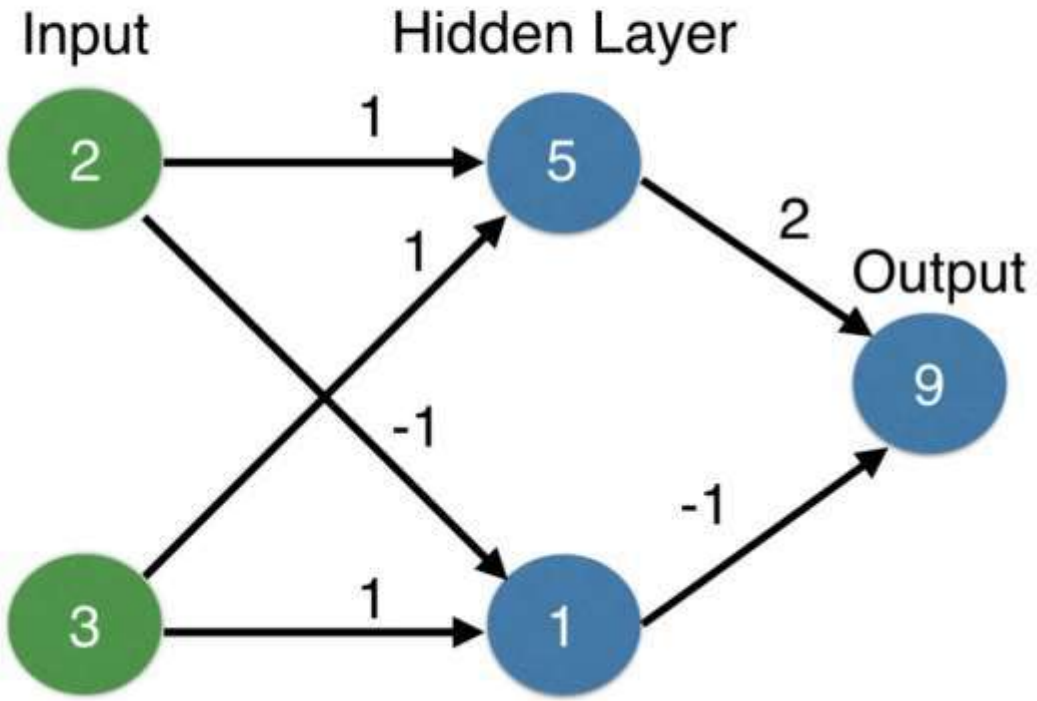
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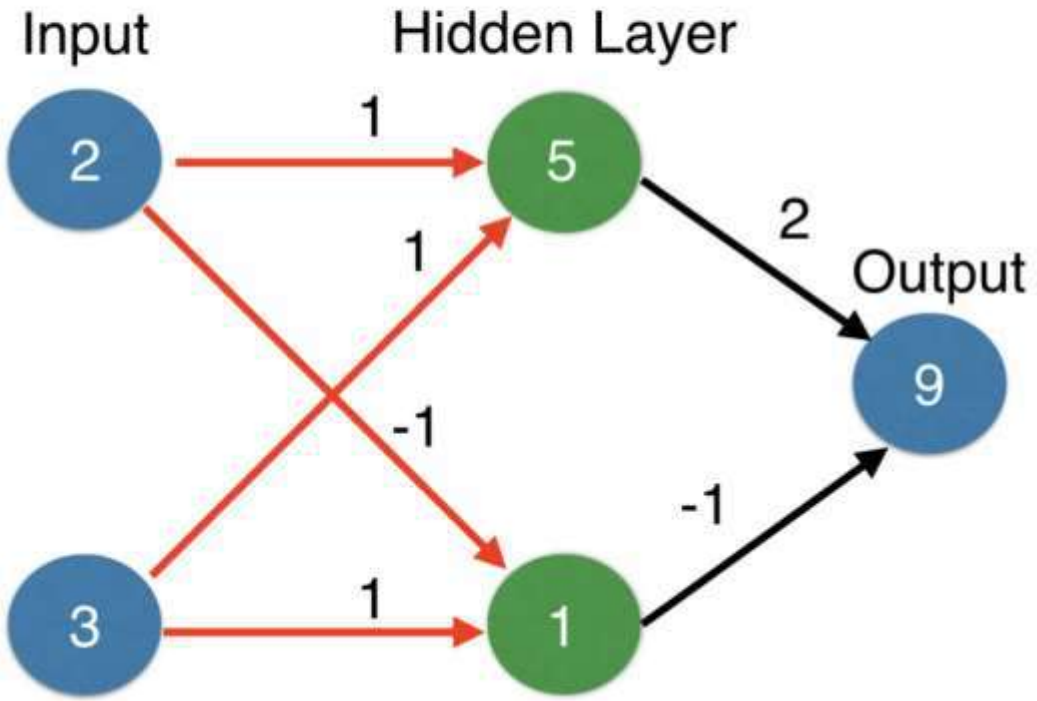
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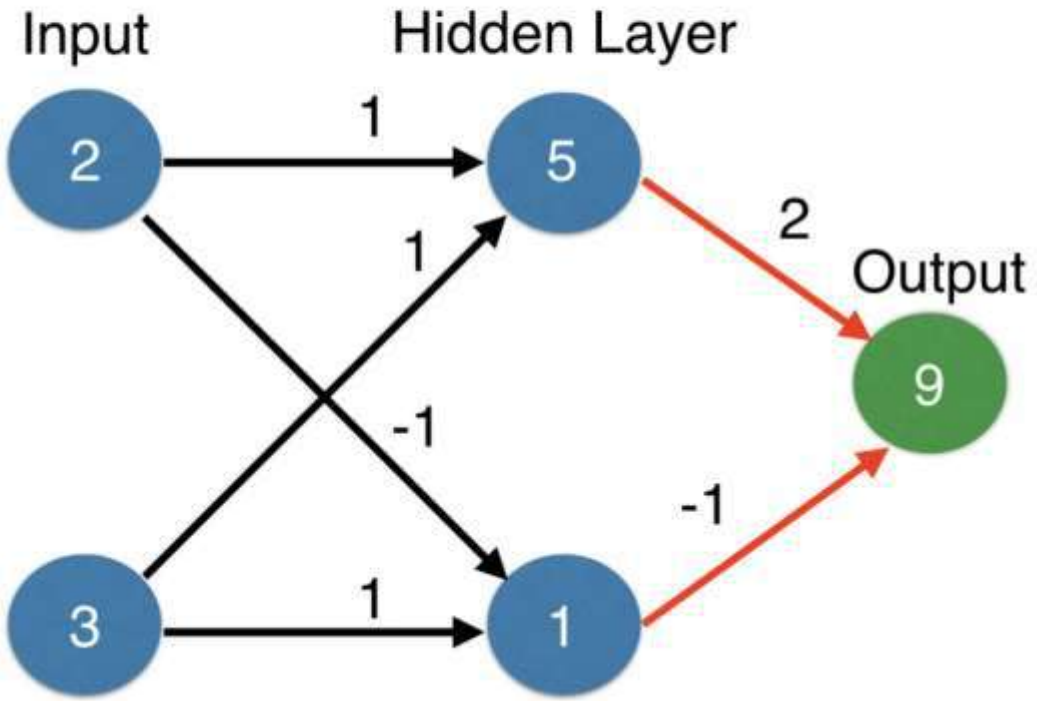
Forward propagation



Forward propagation



Forward propagation



Forward propagation

- Multiply - add process
- Dot product
- Forward propagation for one data point at a time
- Output is the prediction for that data point

Forward propagation code

```
import numpy as np
input_data = np.array([2, 3])
weights = {'node_0': np.array([1, 1]),
           'node_1': np.array([-1, 1]),
           'output': np.array([2, -1])}
node_0_value = (input_data * weights['node_0']).sum()
node_1_value = (input_data * weights['node_1']).sum()
```

Forward propagation code

```
hidden_layer_values = np.array([node_0_value, node_1_value])
print(hidden_layer_values)
```

```
[5, 1]
```

```
output = (hidden_layer_values * weights['output']).sum()
print(output)
```

```
9
```

Let's practice!

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Activation functions

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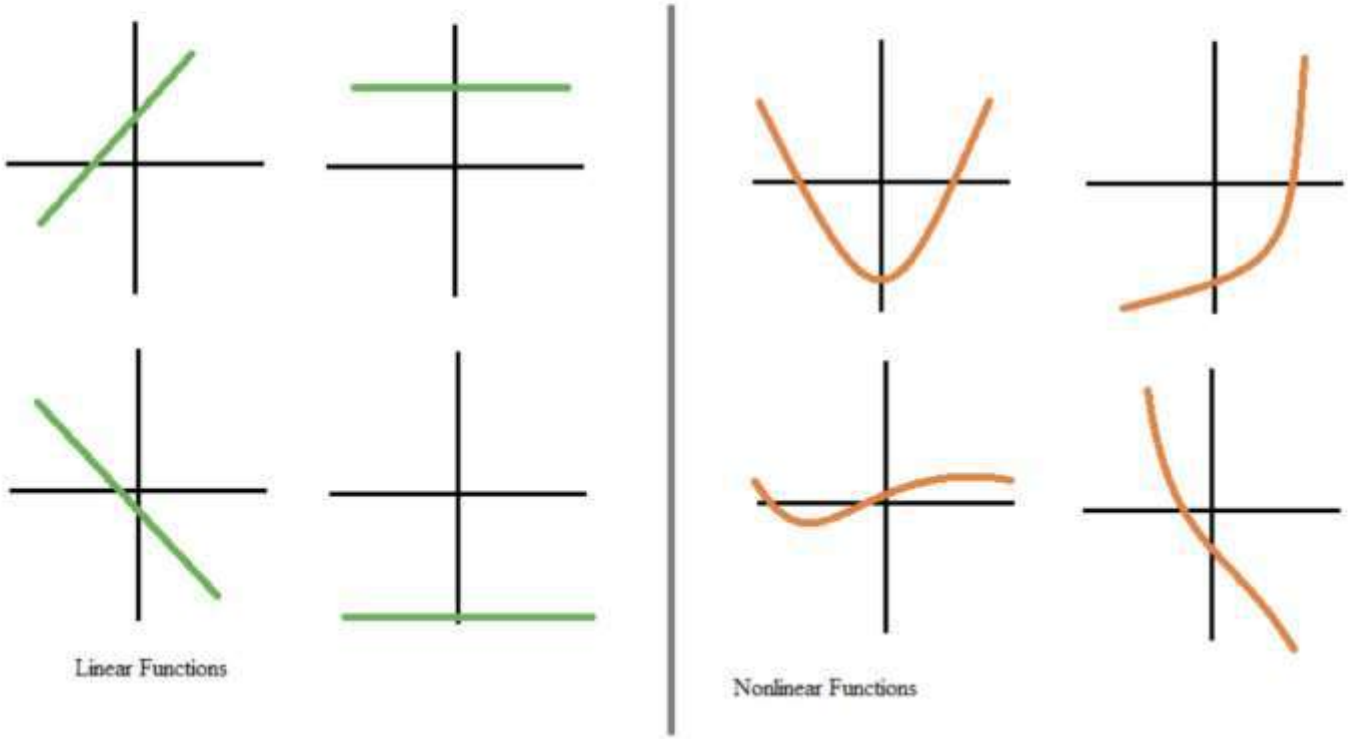


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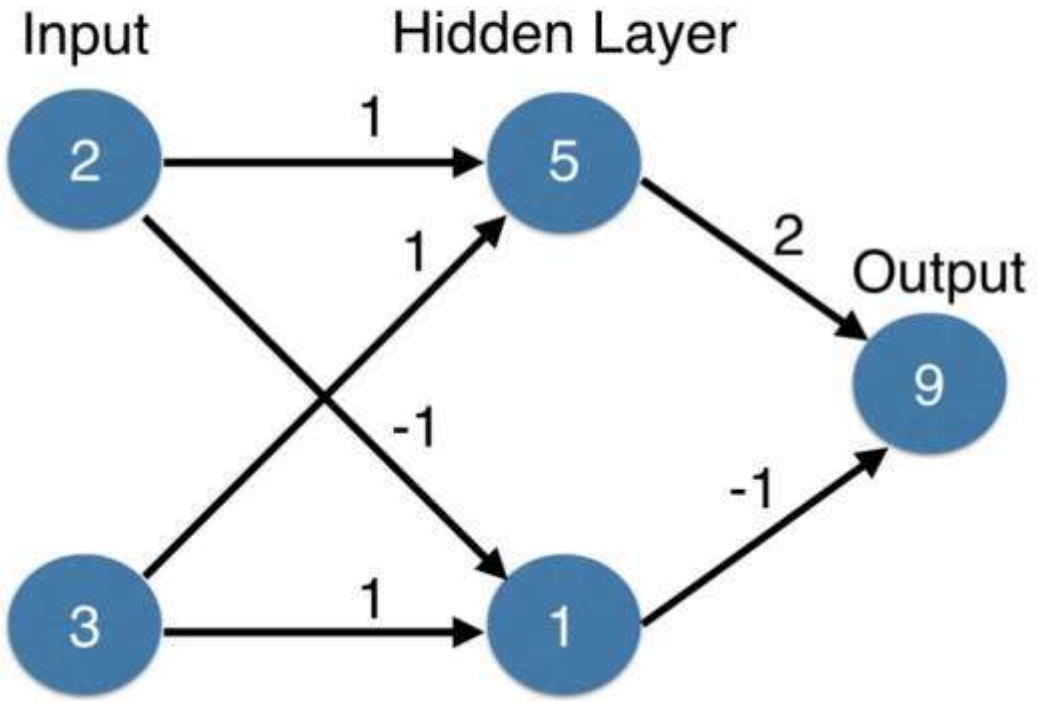
Linear vs. non-linear Functions



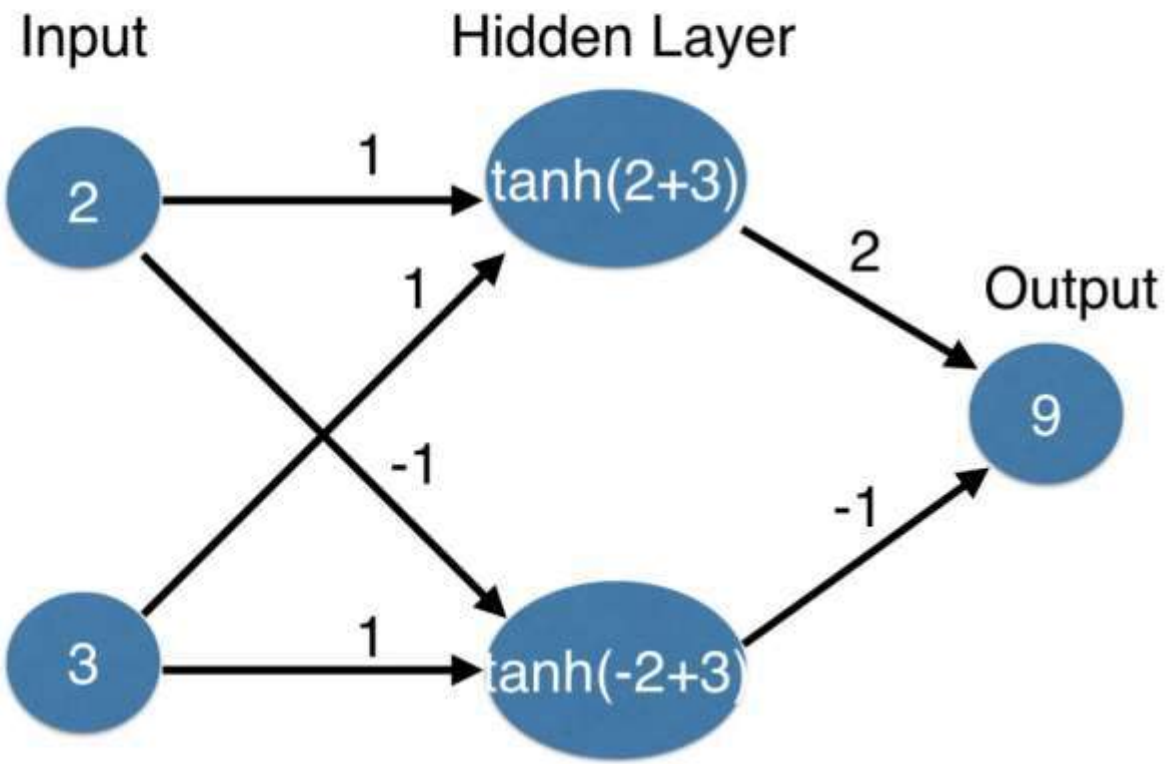
Activation functions

- Applied to node inputs to produce node output

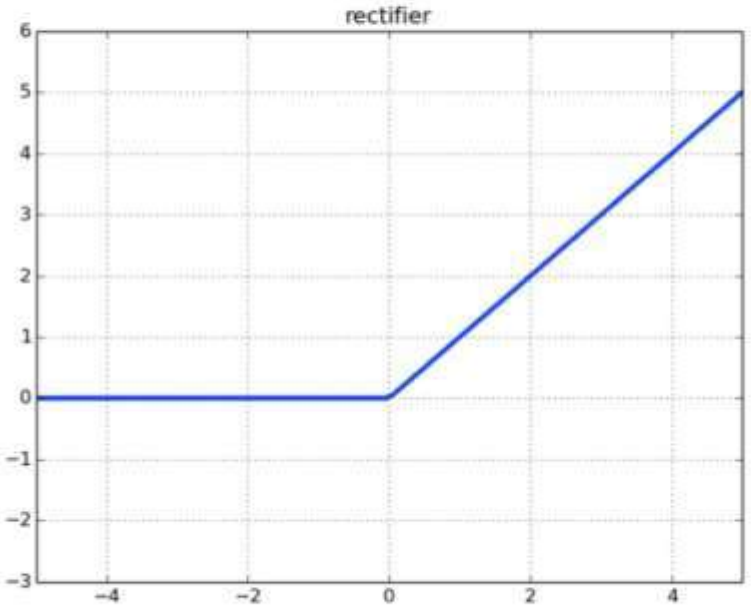
Improving our neural network



Activation functions



ReLU (Rectified Linear Activation)



$$RELU(x) = \begin{cases} 0 \; if \; x < 0 \\ x \; if \; x >= 0 \end{cases}$$

Activation functions

```
import numpy as np
input_data = np.array([-1, 2])
weights = {'node_0': np.array([3, 3]),
               'node_1': np.array([1, 5]),
               'output': np.array([2, -1])}
node_0_input = (input_data * weights['node_0']).sum()
node_0_output = np.tanh(node_0_input)
node_1_input = (input_data * weights['node_1']).sum()
node_1_output = np.tanh(node_1_input)
hidden_layer_outputs = np.array([node_0_output, node_1_output])
output = (hidden_layer_output * weights['output']).sum()
```

```
print(output)
```

```
1.2382242525694254
```

Let's practice!

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Deeper networks

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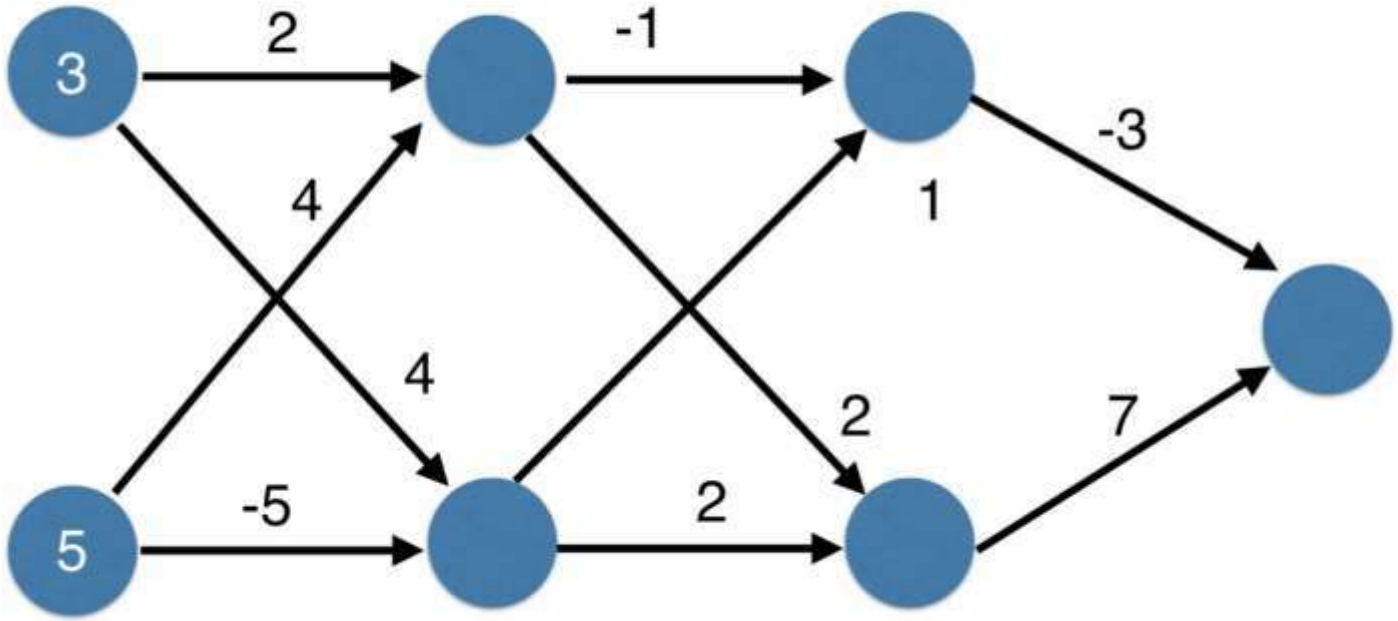


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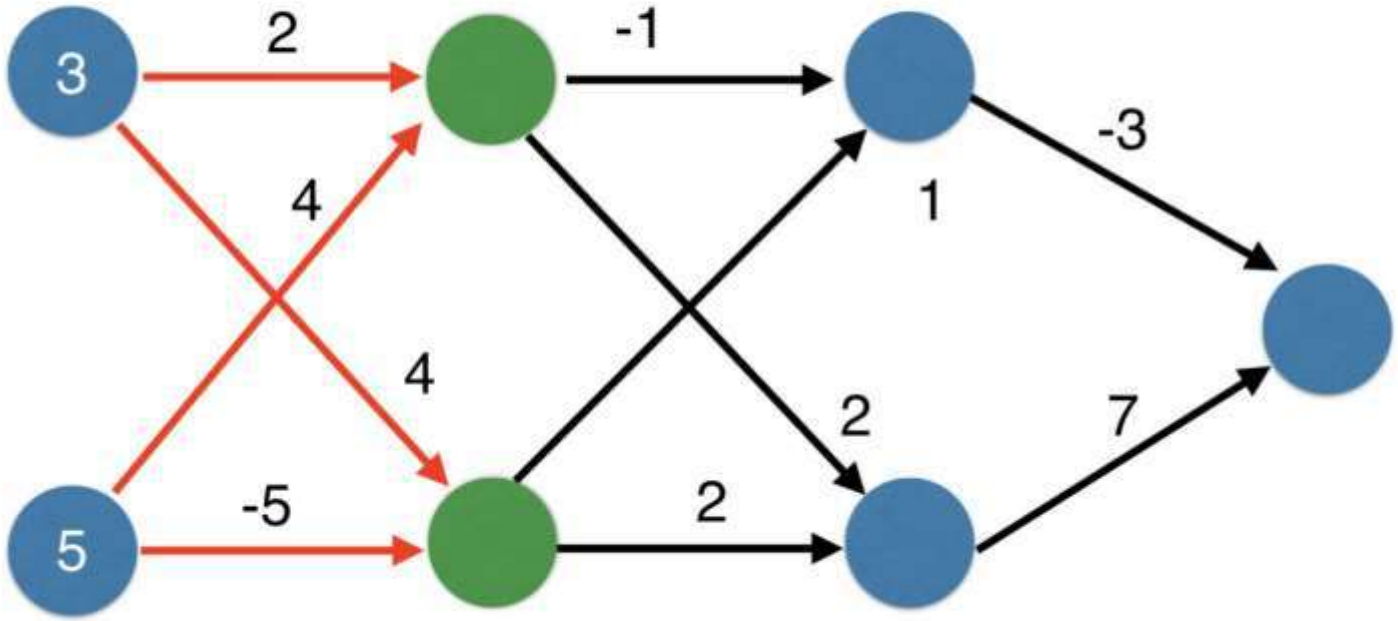
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Multiple hidden layers



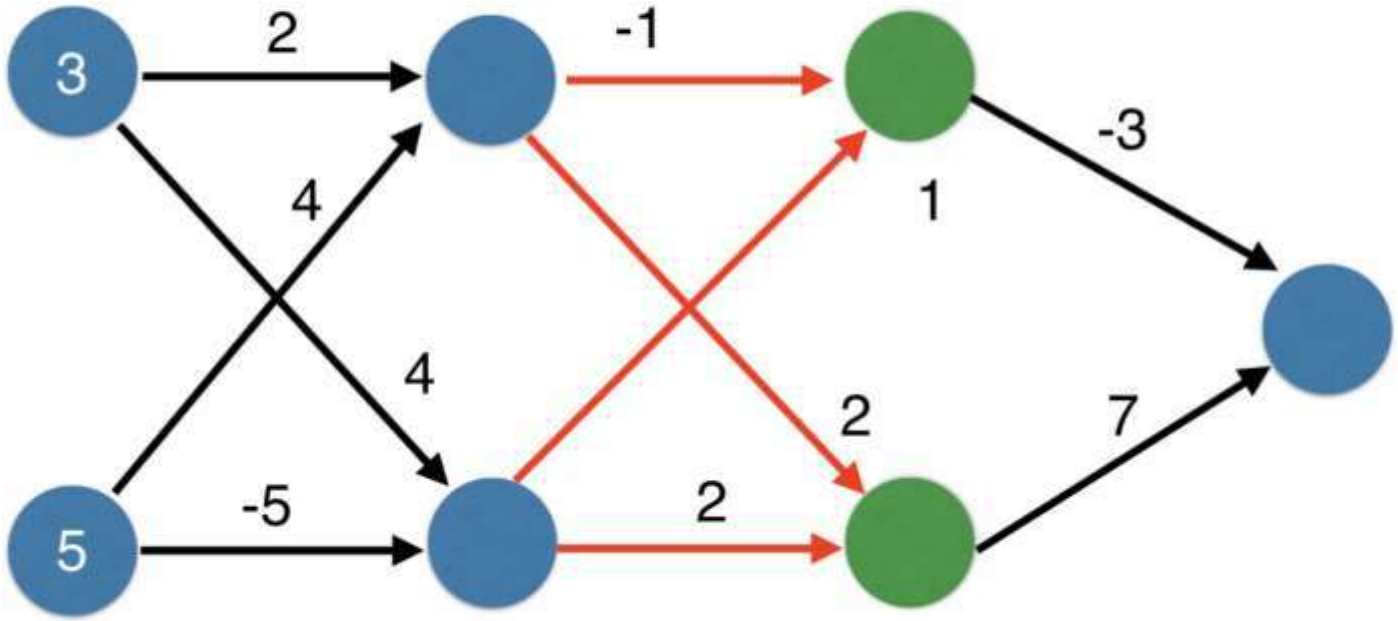
Calculate with ReLU Activation Function

Multiple hidden layers



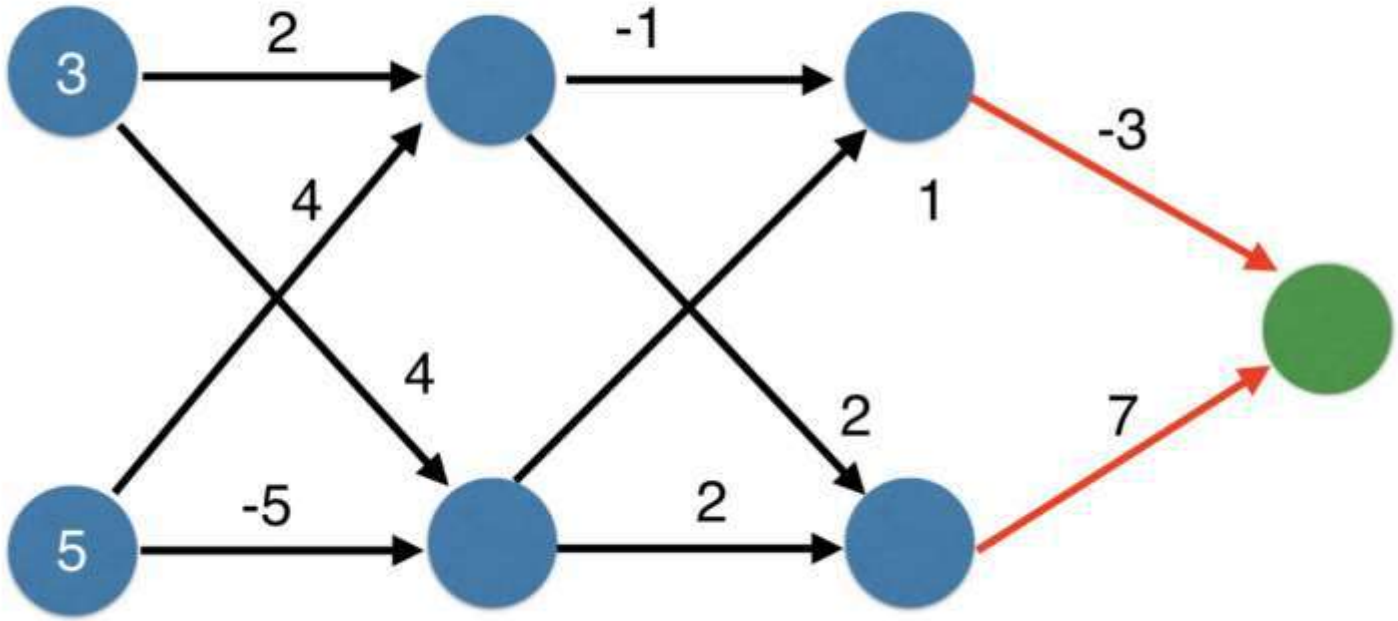
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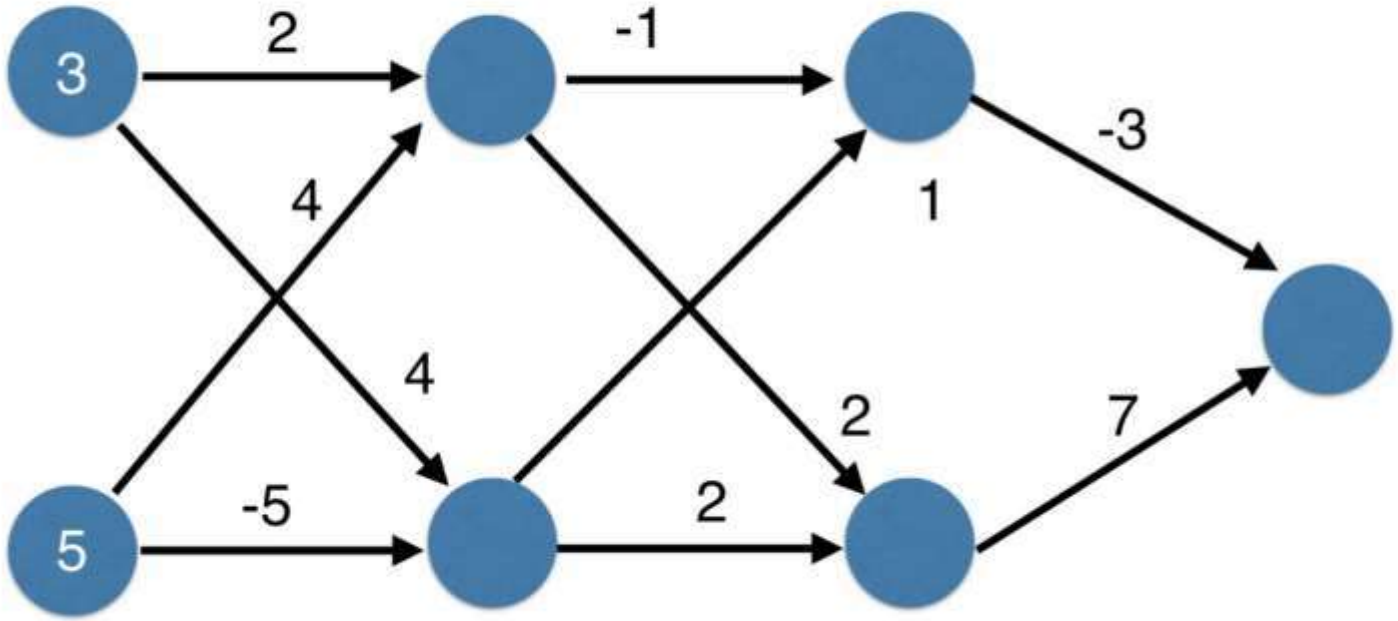
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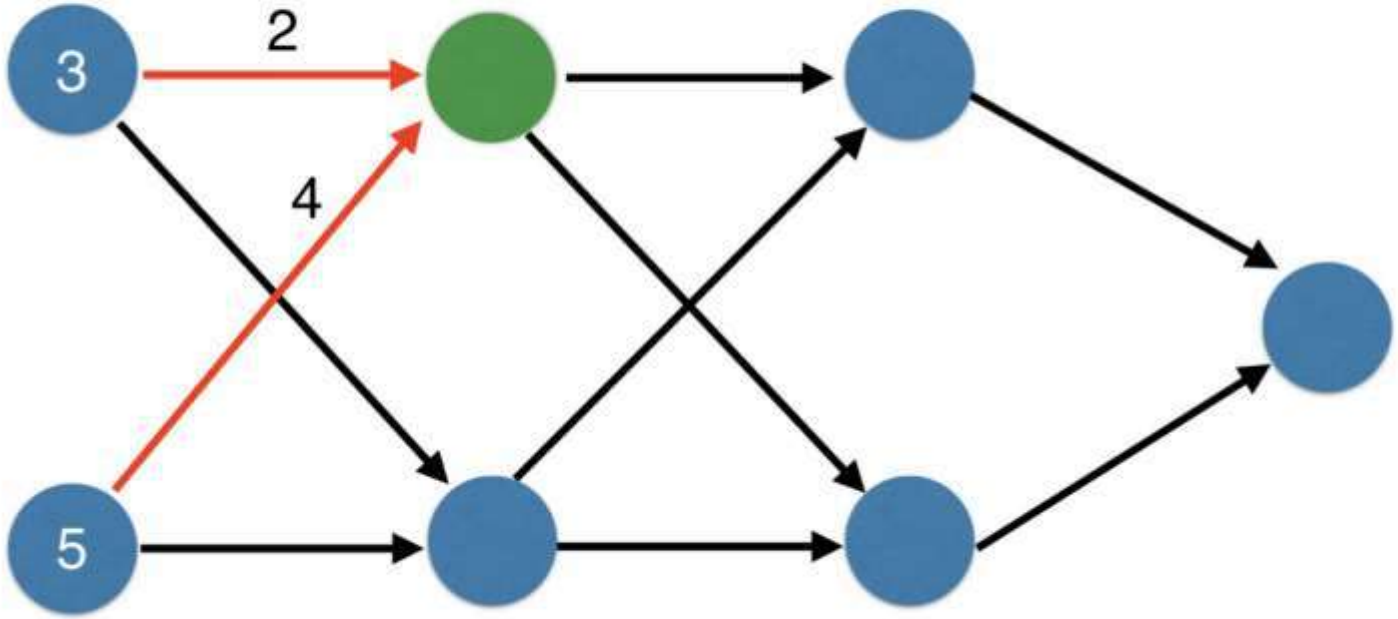
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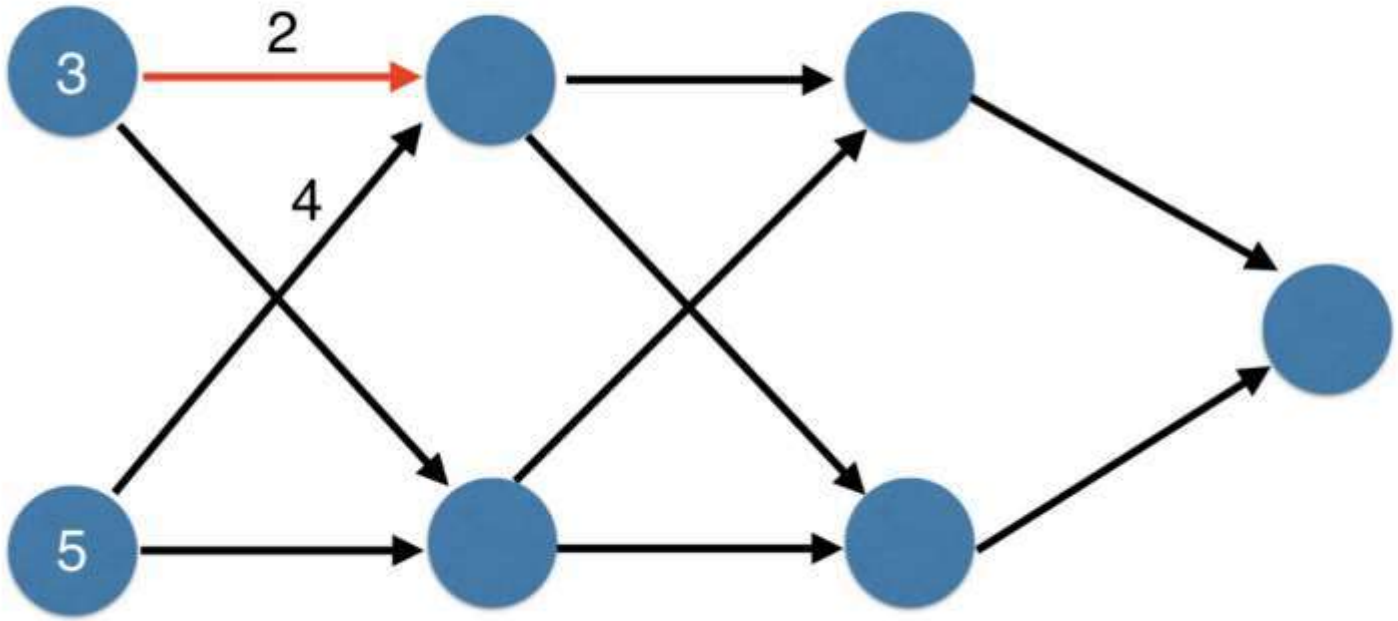
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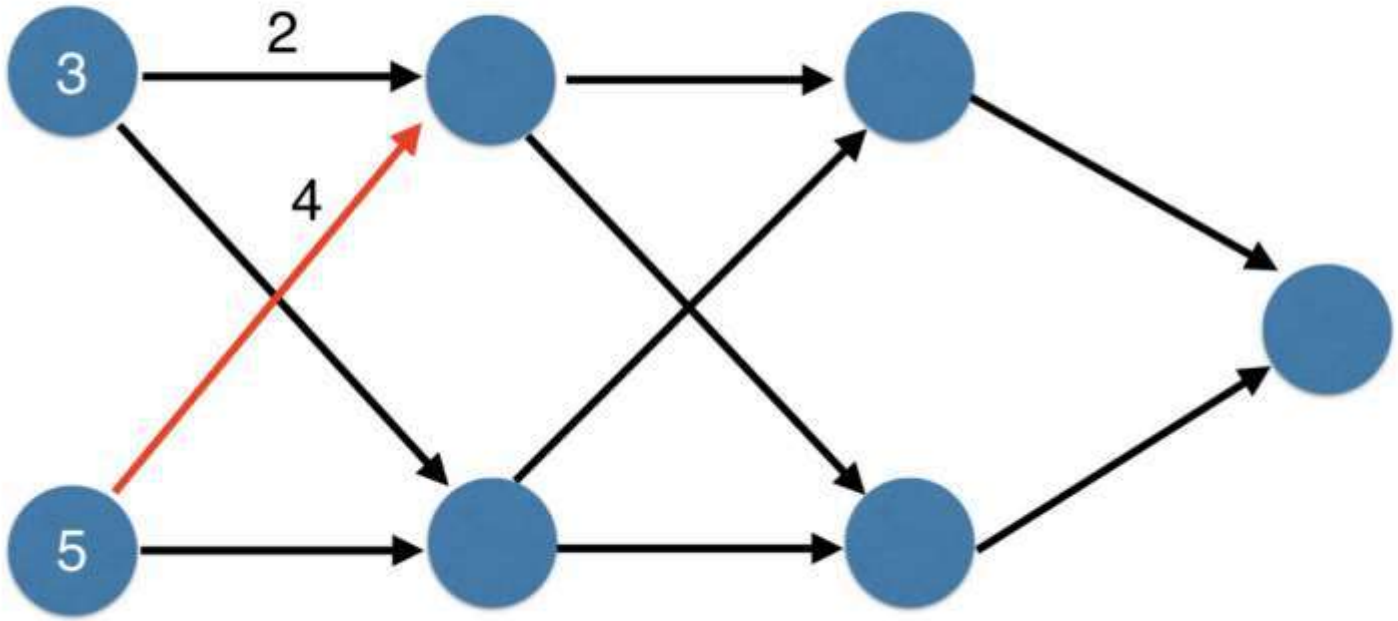
Calculate with ReLU Activation Function

Multiple hidden layers



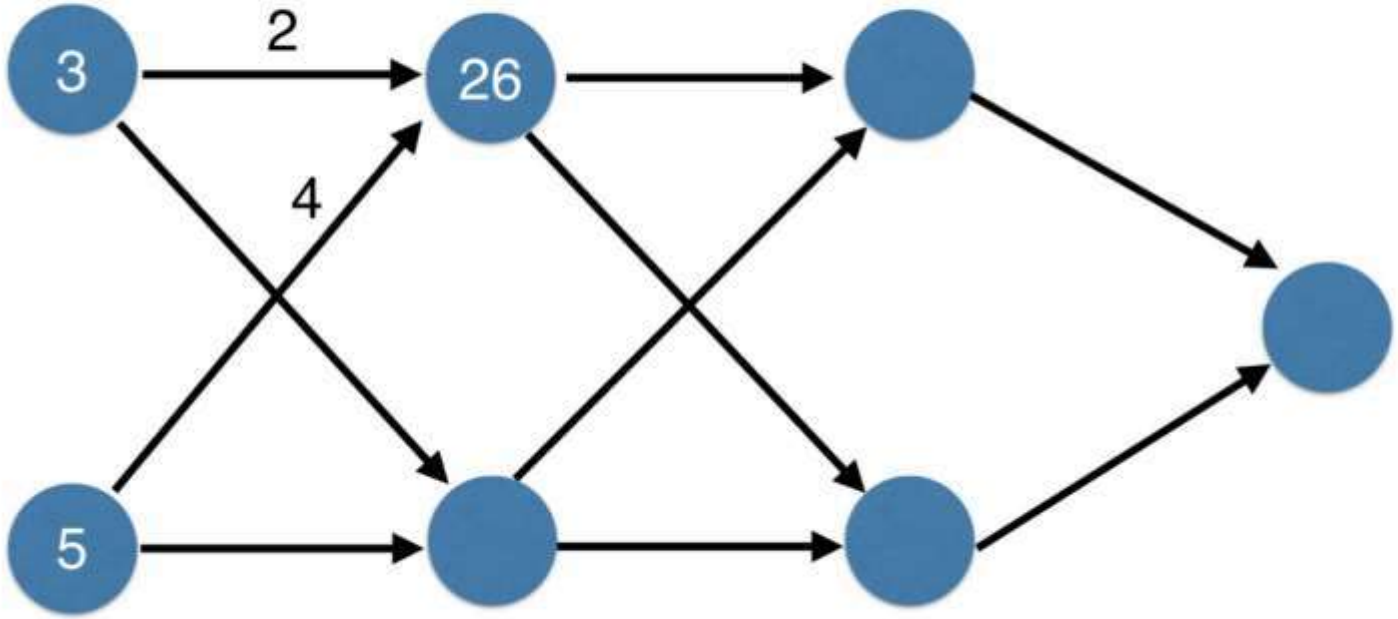
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Multiple hidden layers



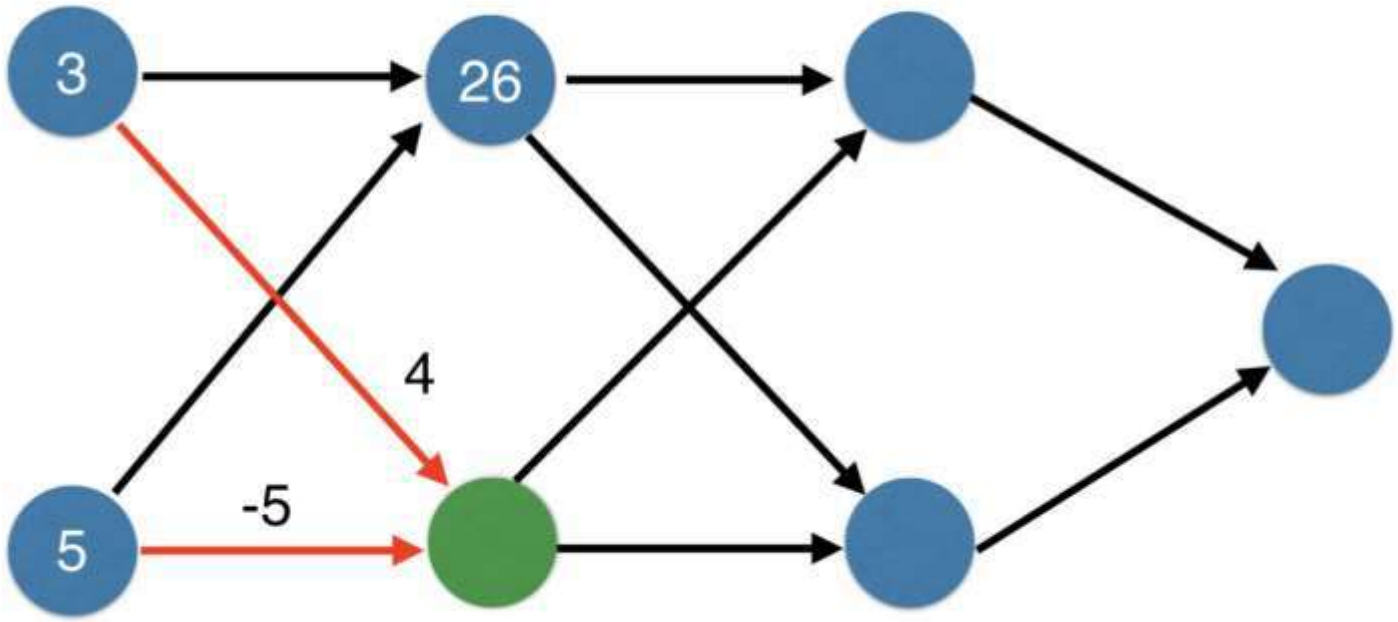
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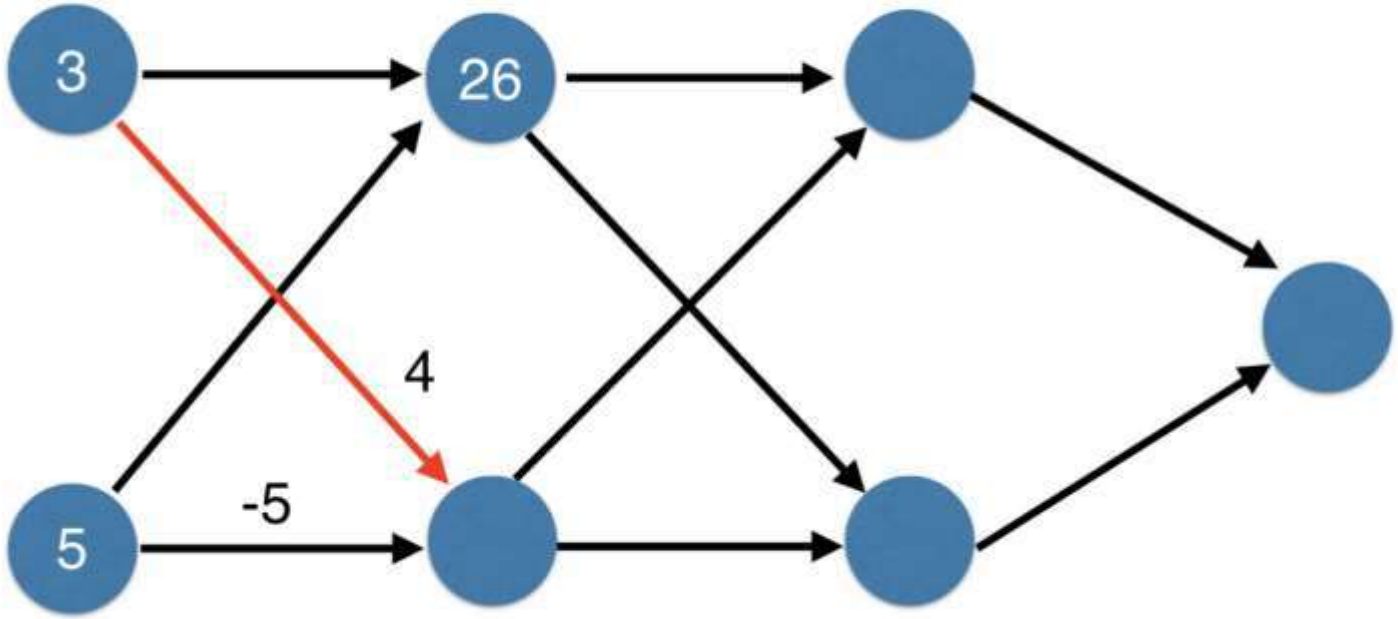
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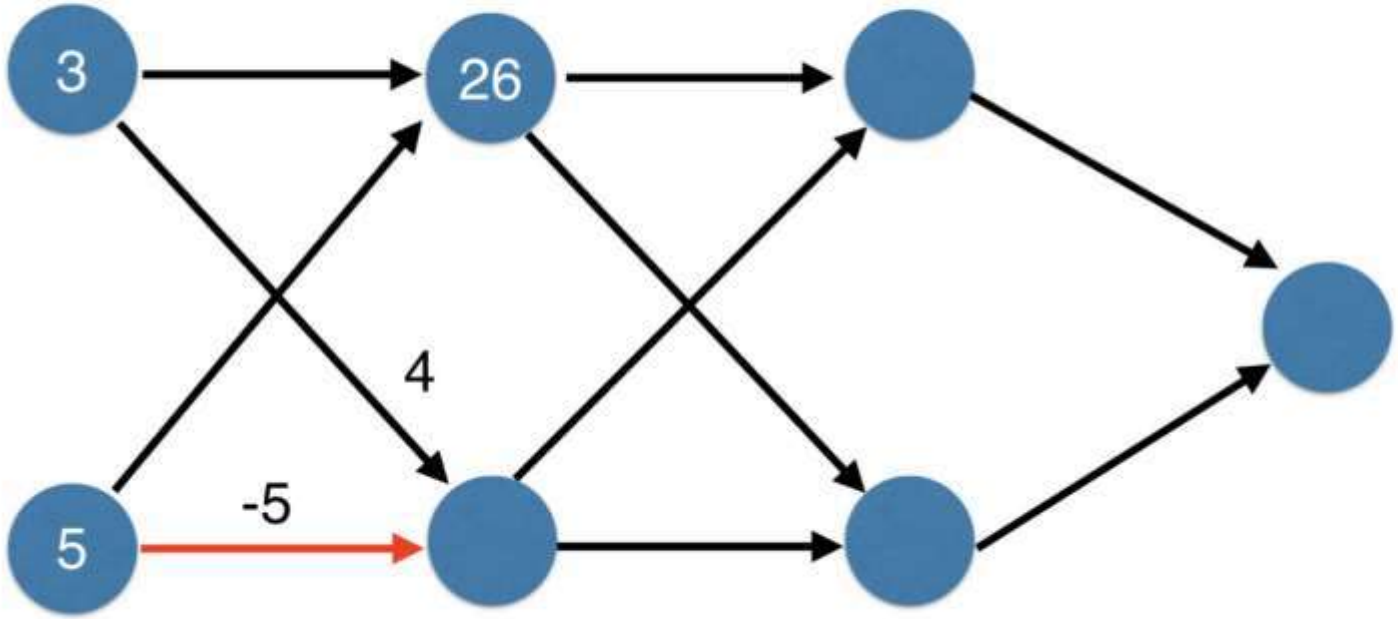
Calculate with ReLU Activation Function

Multiple hidden layers



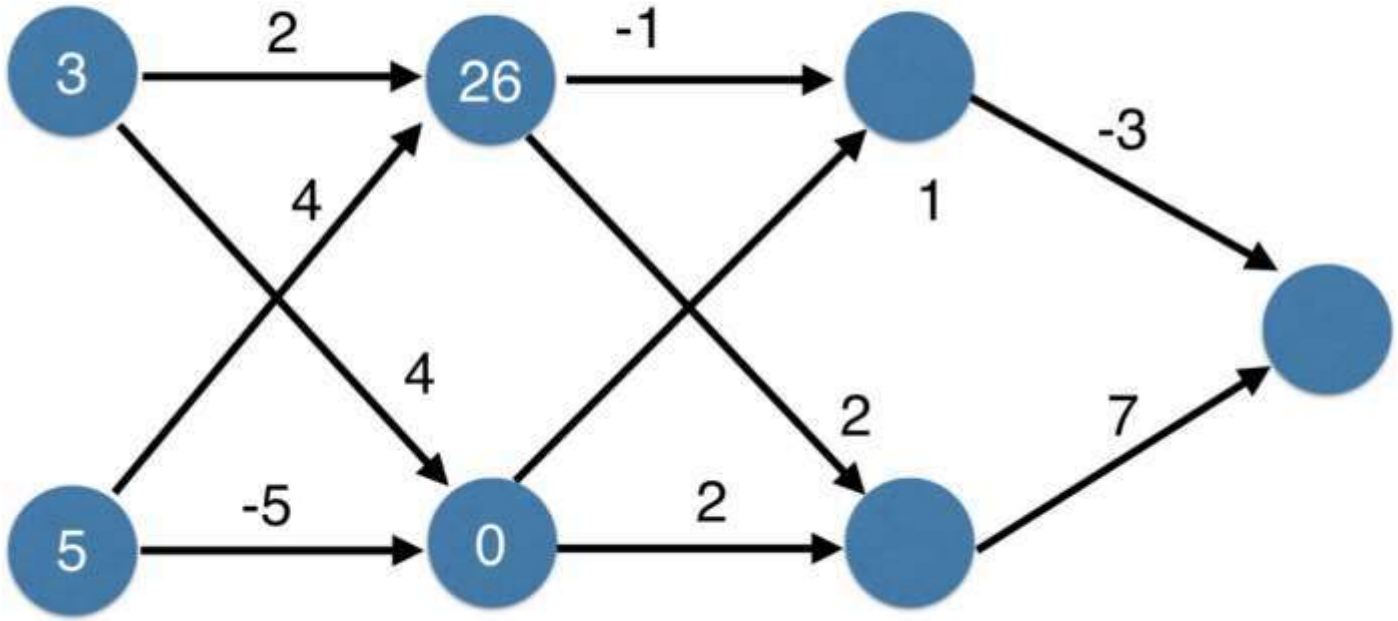
Calculate with ReLU Activation Function

Multiple hidden layers



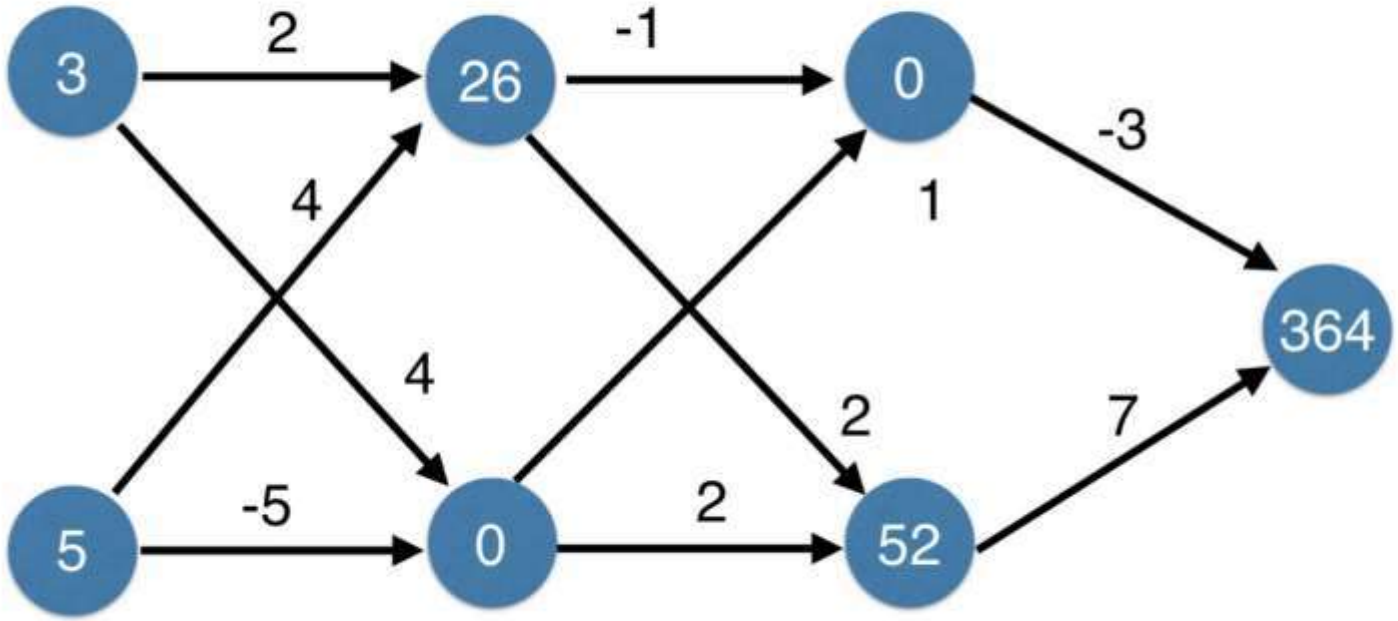
Calculate with ReLU Activation Function

Multiple hidden layers



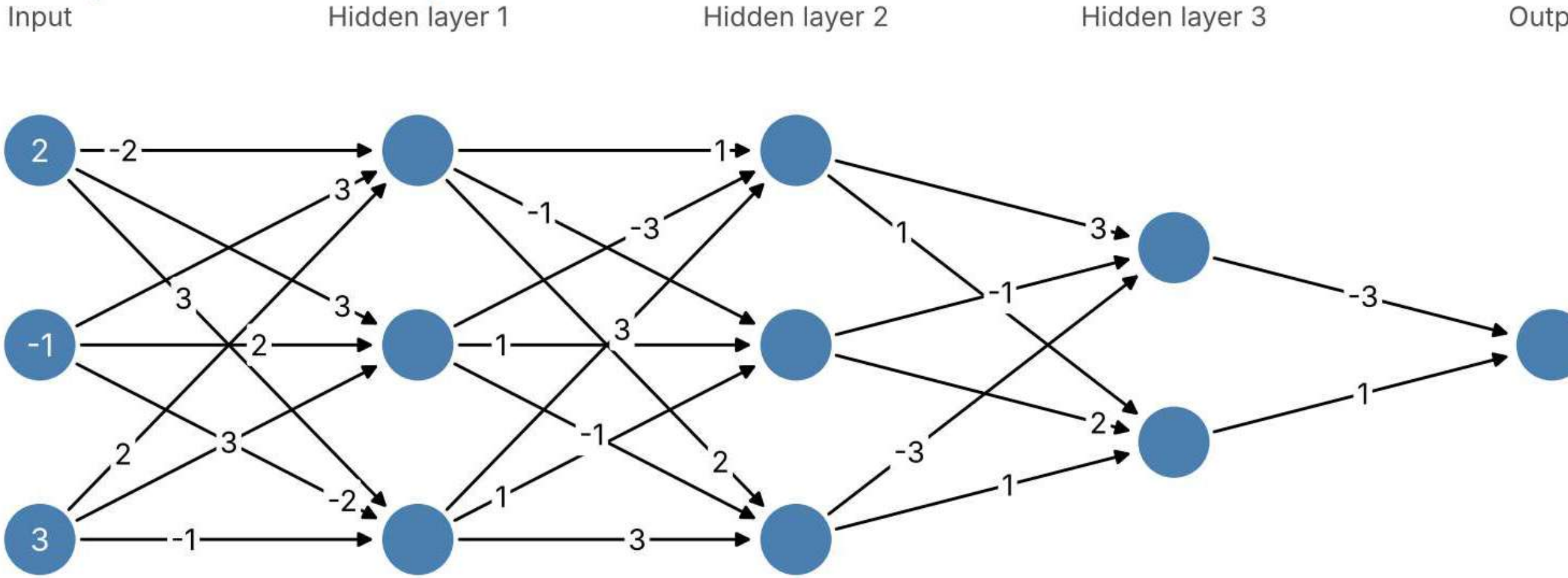
Calculate with ReLU Activation Function

Multiple hidden layers



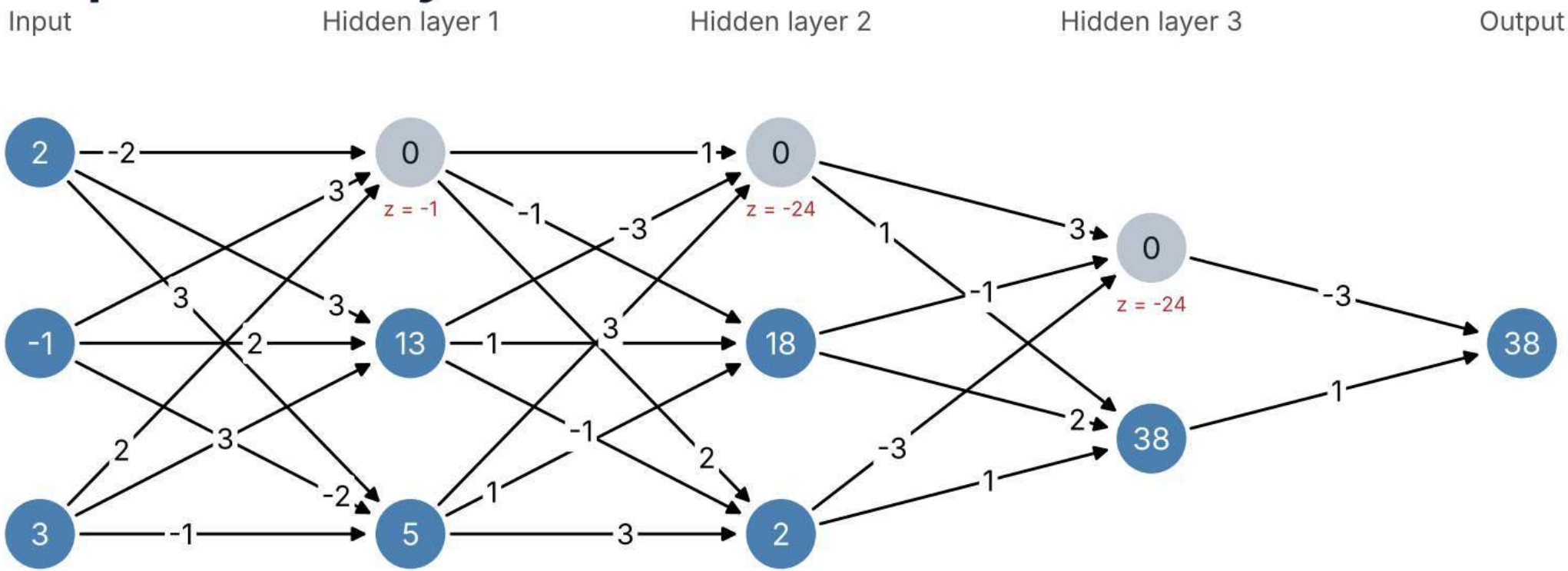
Calculate with ReLU Activation Function

Multiple hidden layers



Calculate with ReLU Activation Function

Multiple hidden layers



Calculate with ReLU Activation Function (grey node: input was negative, ReLU clips it to 0)